

力学系の大域的計算理論入門

17 Feb 2025

集中講義@京大人環

荒井 迅
Zin Arai



Institute of
SCIENCE TOKYO



(Topological) set-oriented computations

Global Computation : keep track of sets, instead of orbits.

- ▶ (Why?) A single point is of measure zero, so its orbit might not reflect dynamically important behaviors.
- ▶ (How?) It is generally impossible to compute a set's image exactly, so we will construct a “rigorous outer approximation” of it and use some math to extract some information on the dynamics.

Remark: In this talk, we only consider “topological computations” that don’t involve measures, weights, etc.

Remark: Some dynamical objects are “uncomputable” even for simple systems such as quadratic maps.

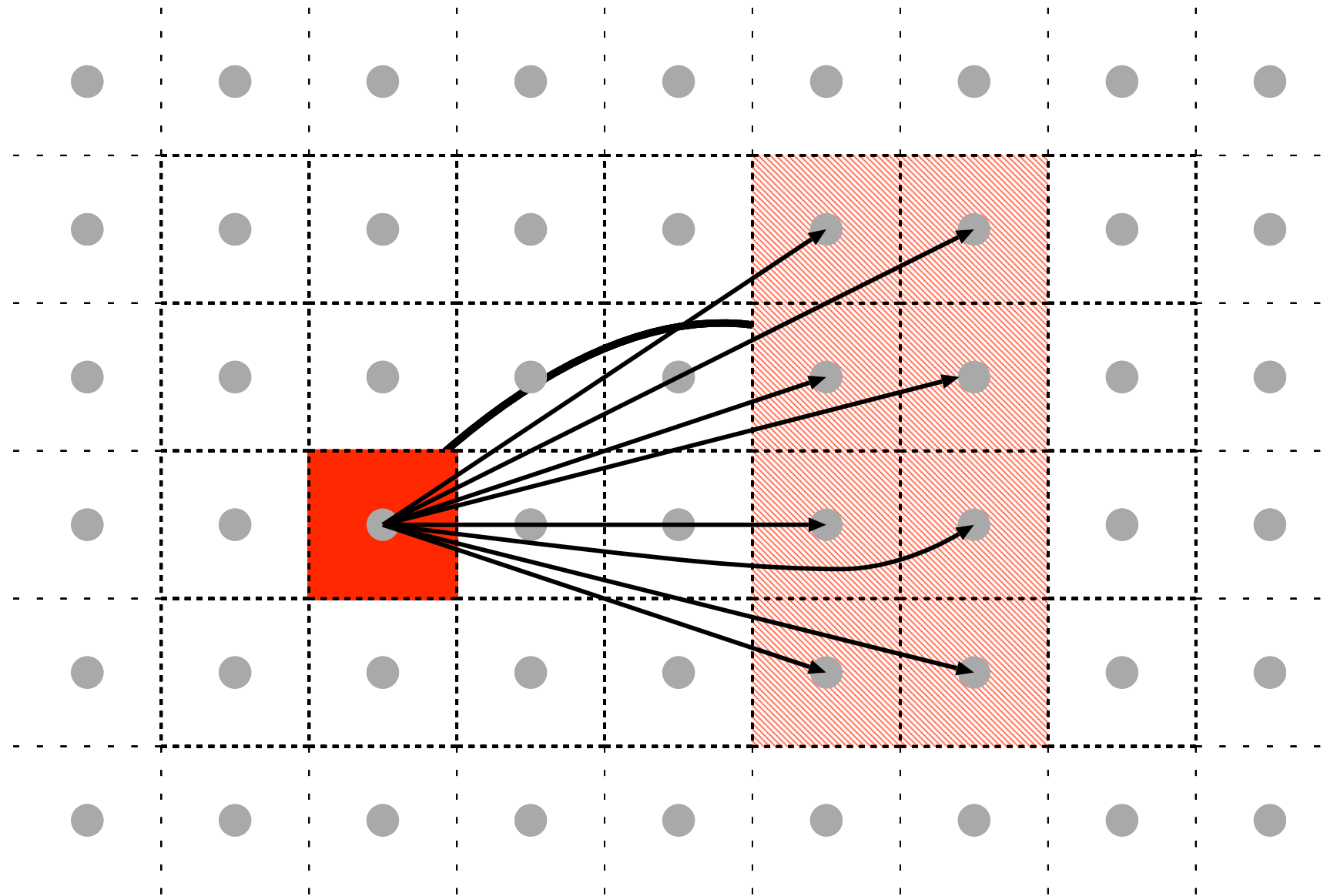
Contents

1. Outer Approximations
2. Application to the verification of uniform hyperbolicity
3. Computational Costs of Global Computations
4. Information theoretic entropy vs topological entropy
5. Computability of Julia Sets

Outer Approximations

The graph $G(f)$

$f : X \rightarrow X$: the dynamics we want to study



Invariant sets

What we want to know:

$$\text{Inv}(f, N) := \{x \in N \mid f^t(x) \in N \text{ for all } t \in \mathbb{Z}\}$$

What we can compute:

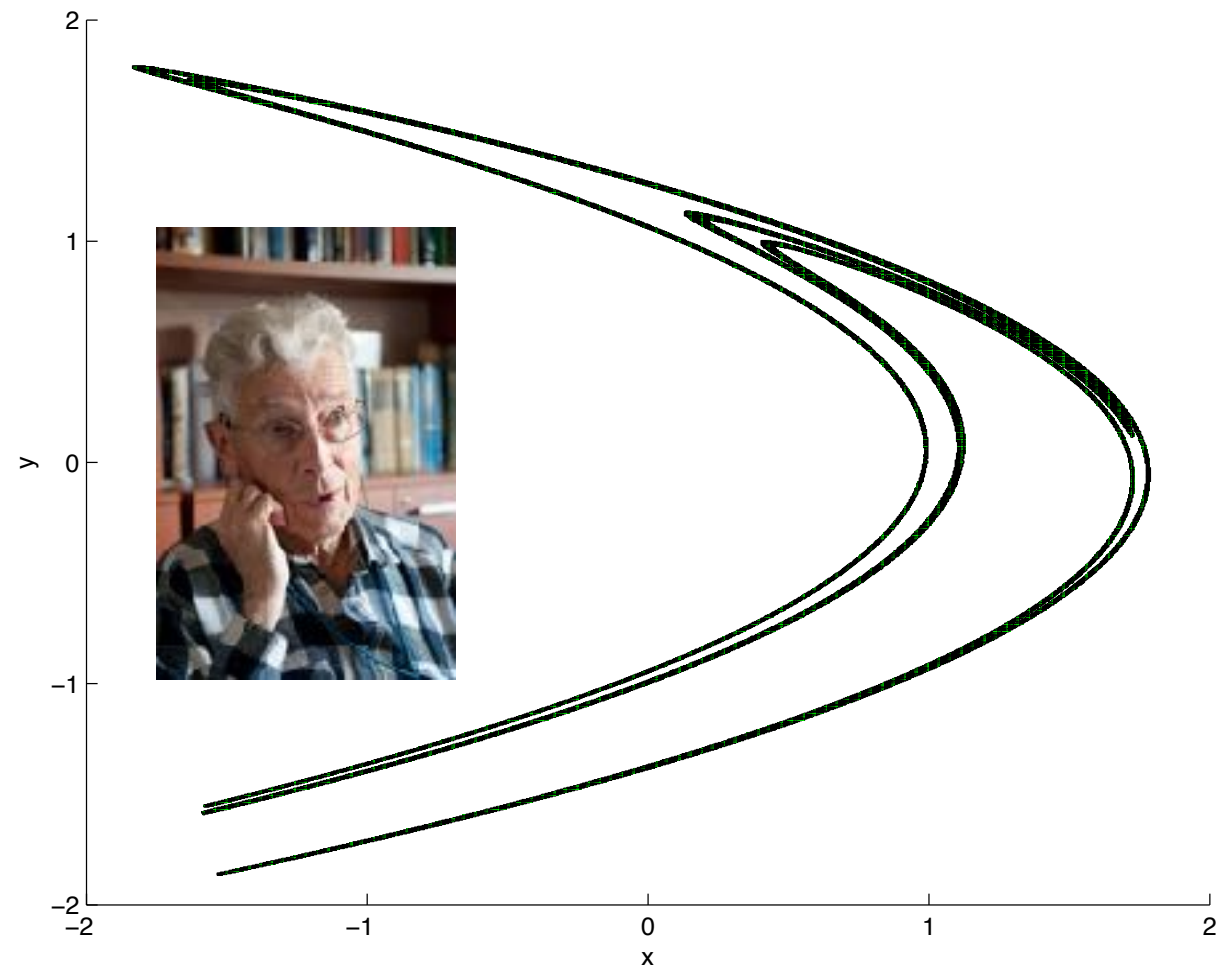
$$\text{Inv } G(f) := \{v \in G(f) \mid \exists \text{ bi-infinitely long path through } v\}$$

Theorem. $\text{Inv}(f, N) \subset |\text{Inv } G(f)|$

The Hénon map

The Hénon map:

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} a - x^2 + by \\ x \end{pmatrix}$$

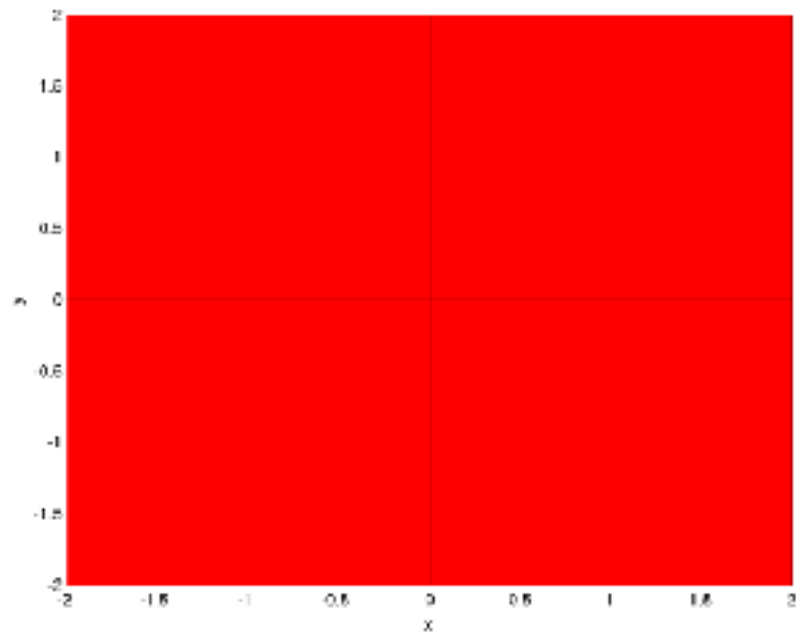


Michel Hénon observed a “strange” attractor
at $a = 1.4$, $b = 0.3$

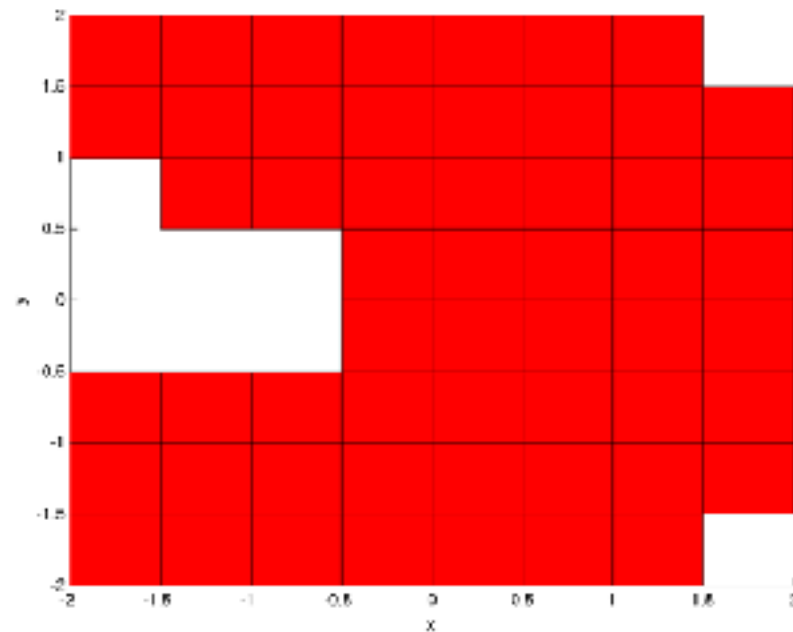
The map is:

- * a diffeomorphism if b is non-zero.
- * conjugate to the quadratic map if $b = 0$.
- * only nontrivial 2-dim polynomial diffeomorphism of deg. 2.
- * a role model of nonlinear dynamics in dim. higher than 2.

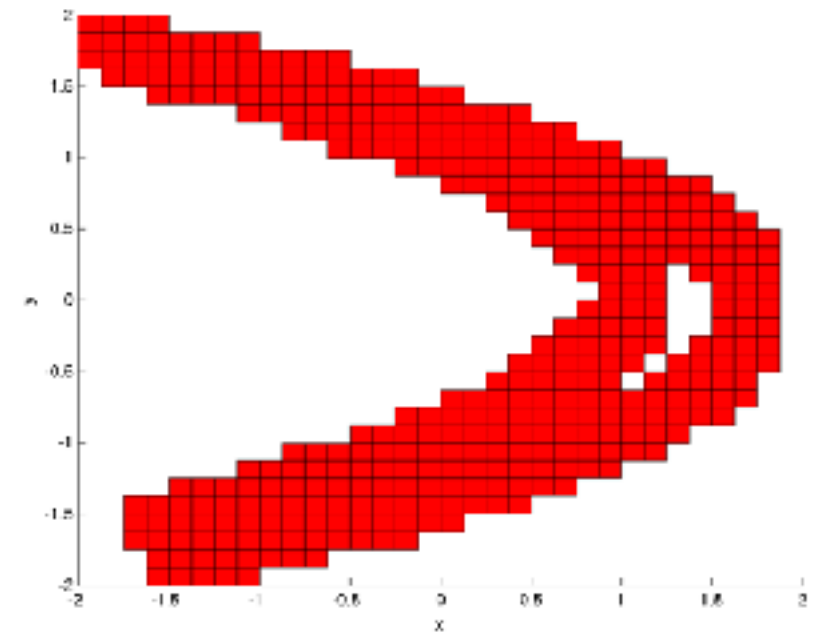
Invariant set of the Hénon map



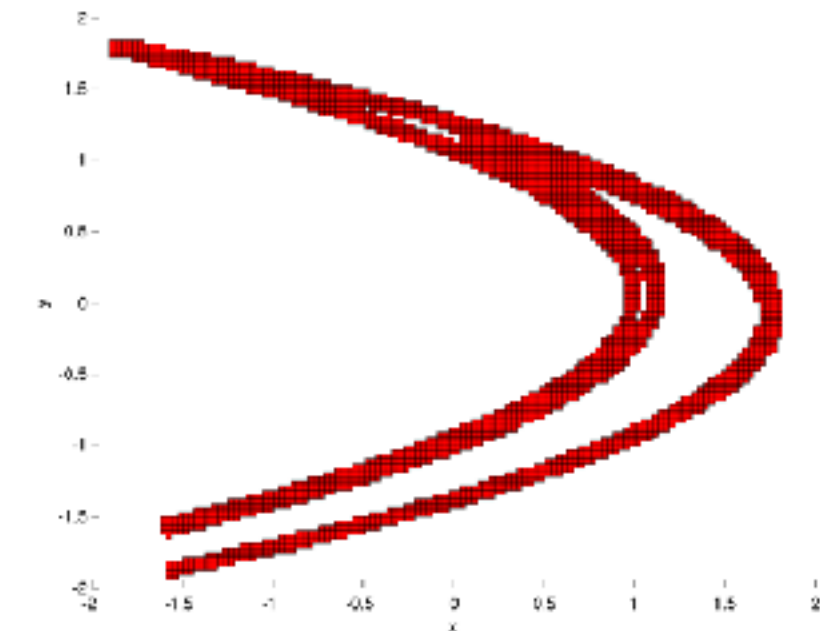
depth 2



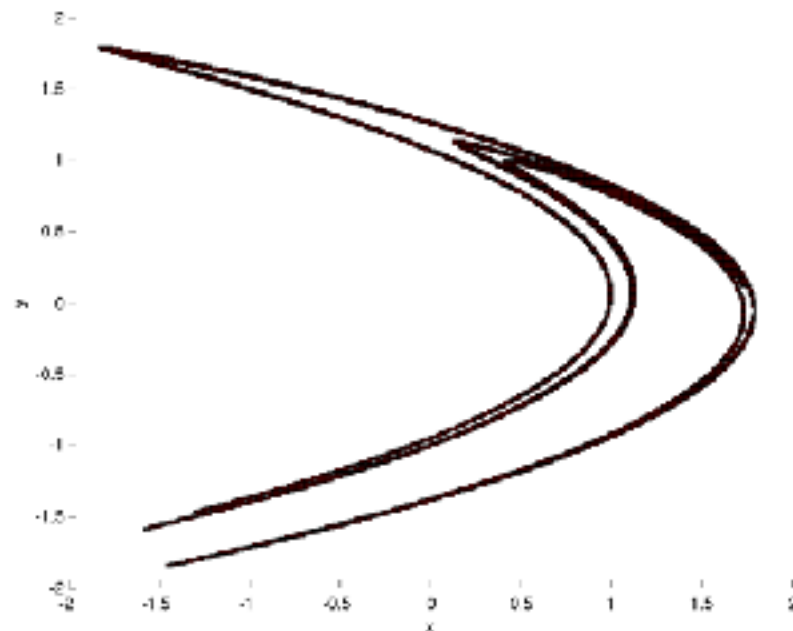
depth 6



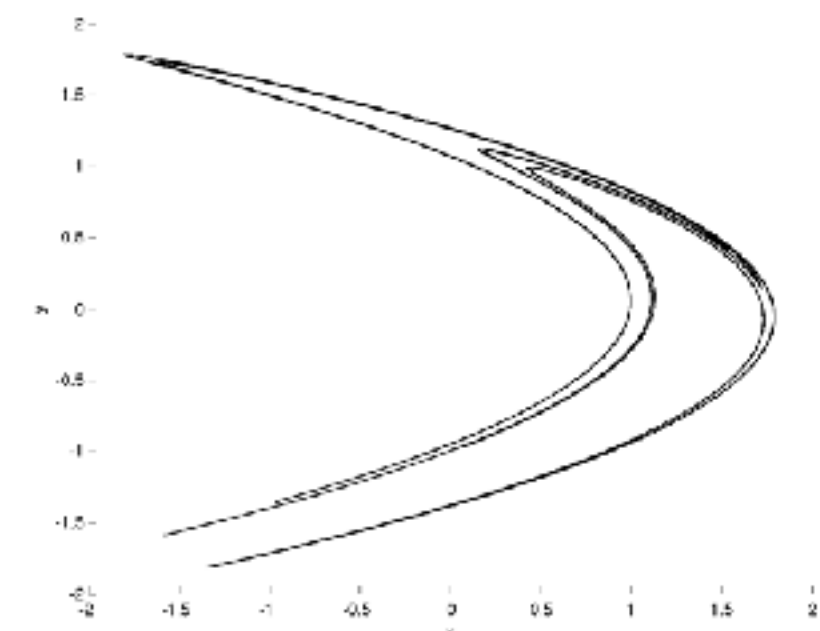
depth 10



depth 14

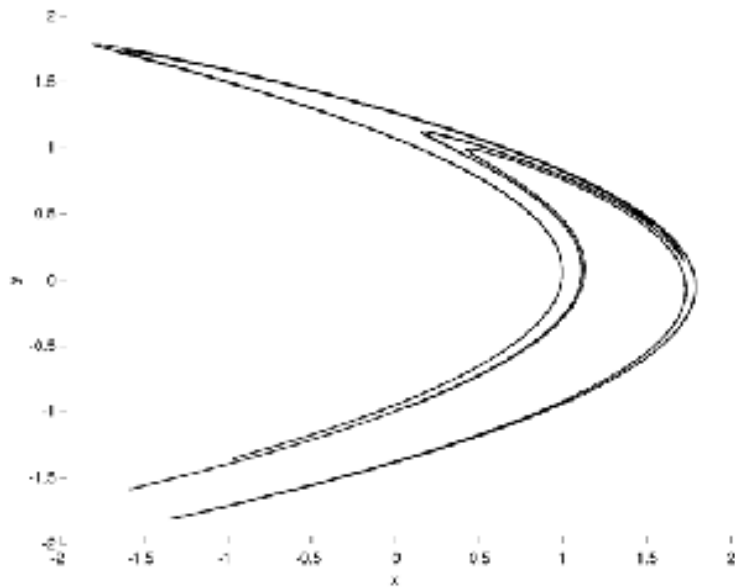


depth 18



depth 22

Non-existence and existence



Since we are using rigorous outer approximations, this computation immediately **proves** that there is no invariant orbit outside these boxes.

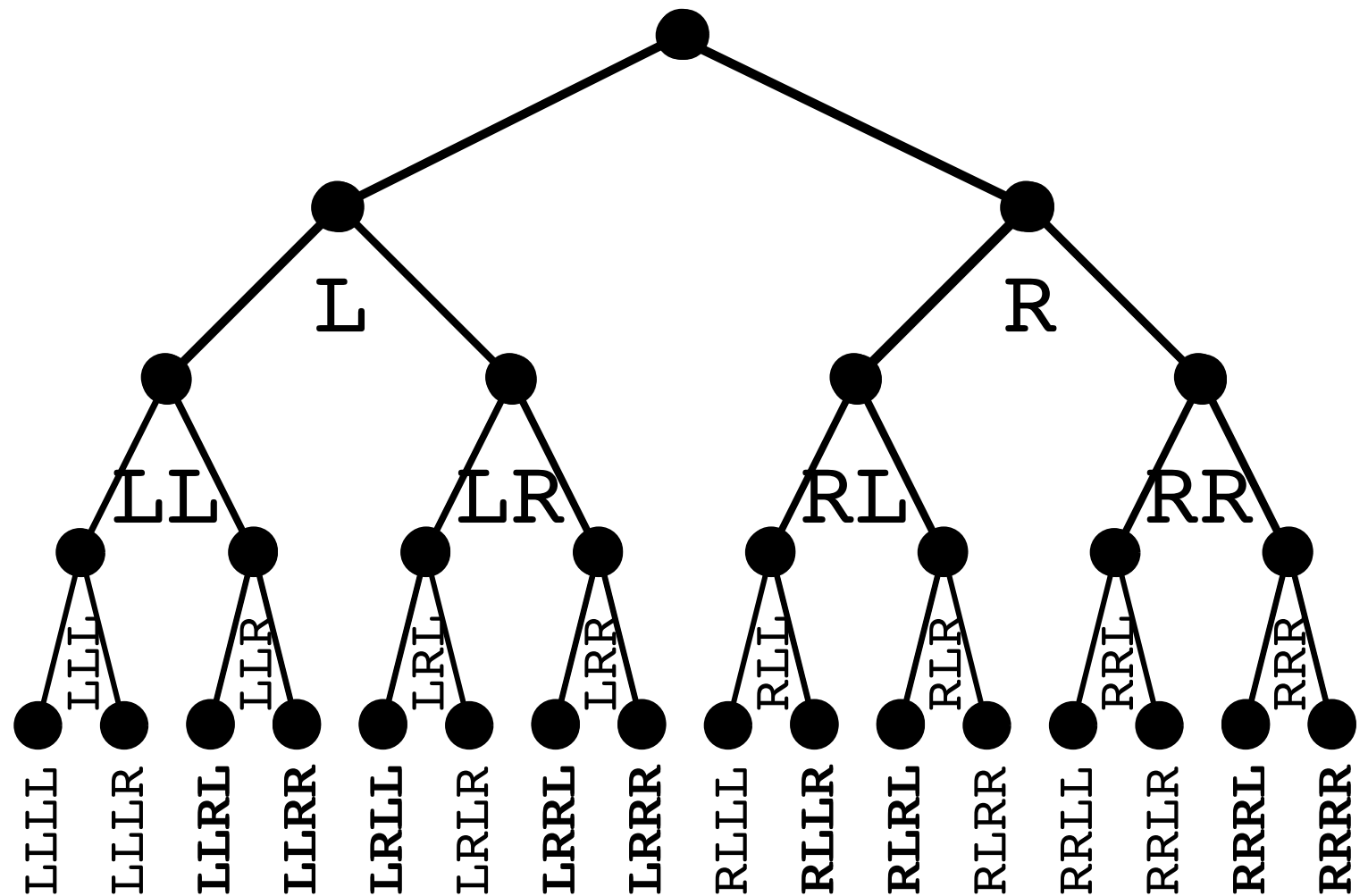
Non-existence: Outer approximations are very useful to prove the non-existence of some dynamical objects.

Existence: To conclude the existence of something, we need to use a mathematical tool such as:

- ▶ Interval Newton/Krawczyk methods
- ▶ the Conley index theory (fixed point theorem)
- ▶ the rigidity of holomorphic maps

Binary Tree

In standard implementations, cubical spaces are represented by binary trees

[illegible]

GAIO: Global Analysis of Invariant Objects

<https://github.com/gaioguy/GAIO>

Dellnitz, M.; Froyland, G.; Junge, O.: The algorithms behind GAIO - Set oriented numerical methods for dynamical systems, *in: B. Fiedler (ed.): Ergodic theory, analysis, and efficient simulation of dynamical systems*, Springer, 2001.

- ▶ The core of GAIO is a binary tree that maintains subdivisions of the phase space into small boxes.
- ▶ It seems that everyone has their own version of GAIO.
- ▶ I have partially implemented a Julia version of GAIO which will hopefully be available soon.

Application to the verification of
uniform hyperbolicity

Is it really strange?

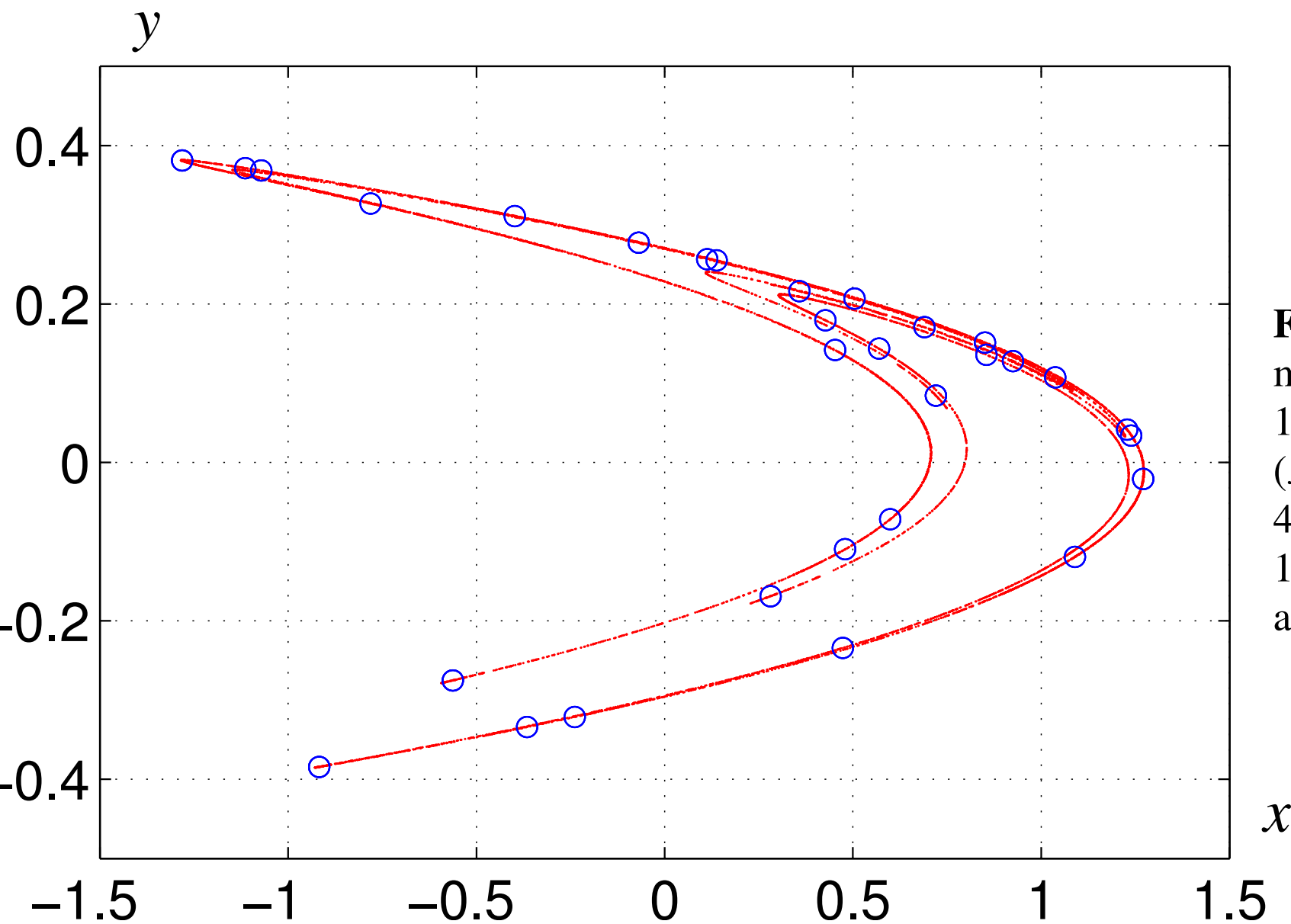
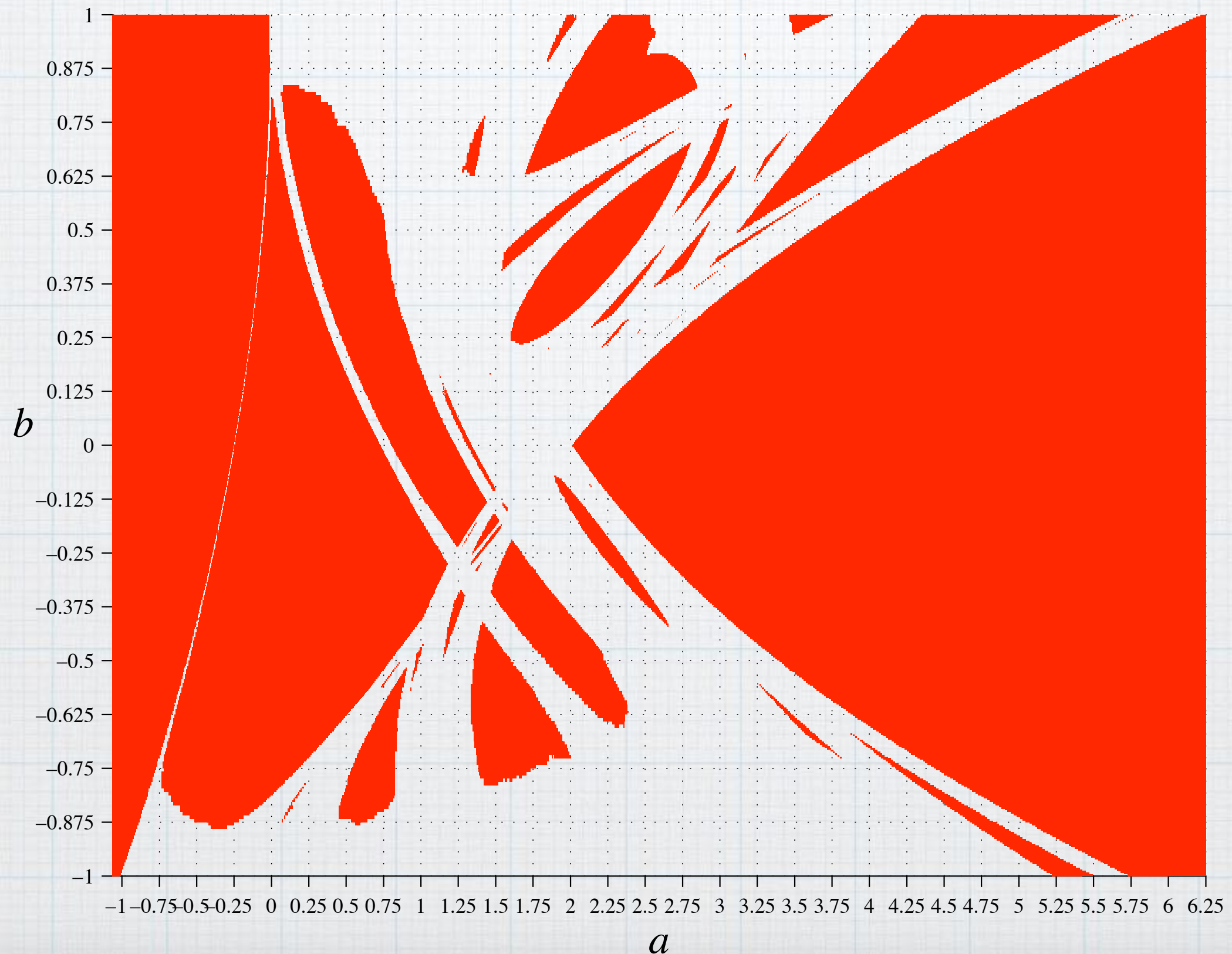


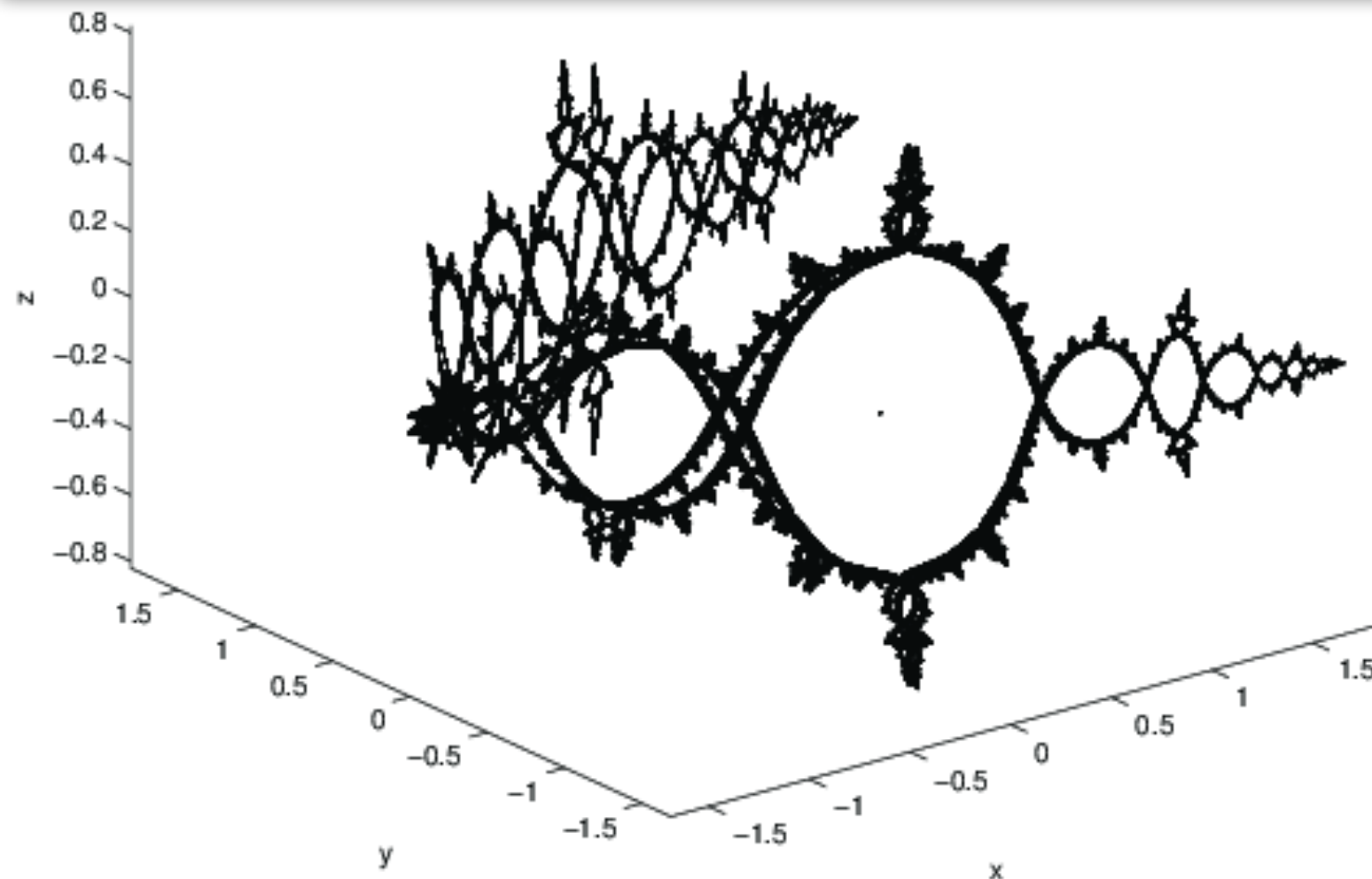
Figure 2. Trajectory of the Hénon map computed in double-precision, $a = 1.4000000009849371$, $b = 0.3000000019143266$, $(x_0, y_0) = (0, 0)$, 10000 points after skipping 4×10^{11} iterations are plotted using red dots, 10000 points after skipping 5×10^{11} iterations are plotted using blue circles.

At $a = 1.4000000009849371$, $b = 0.3000000019143266$, the Hénon map looks like a “strange attractor”, but is just a periodic sink (by Zbigniew Galias, doi:10.1088/1742-6596/692/1/012001).

Theorem (ZA) If (a, b) is in the red region, then the chain recurrent set of $H_{a,b}$ is uniformly hyperbolicity.



Example: Complex Hénon Map



The Julia set of
the complex Hénon map
 $(x, y) \mapsto (x^2 + c - ay, x)$
 $a = 0.15, c = -1.1875$

Theorem. This Julia set is uniformly hyperbolic.

To prove this, we need to work in $T\mathbb{C}^2 \cong \mathbb{R}^8$

- * The Julia set is like a hyperbolic Solenoid
- * There are 4 expanding directions

Linear Maps in 1-dim

Let $M = \mathbb{R}$ and $f(x) = \lambda x$.

The differential $D_x f$ is the same as f itself, that is, $D_x f(v) = \lambda v$.

The origin $0 \in \mathbb{R}$ is a fixed point of f and the differential at 0

$$D_0 f(v) = \lambda v$$

has the eigenvalue λ . Therefore,

0 is a hyperbolic fixed point of $f \iff |\lambda| \neq 1$.

$$|\lambda| < 1 \iff f^n(x) \rightarrow 0 \text{ for all } x \in \mathbb{R}$$

$$|\lambda| > 1 \iff f^n(x) \rightarrow \infty \text{ for all } x \in \mathbb{R}$$

Linear Maps in 2-dim

Let $M = \mathbb{R}^2$ and f is defined by 2×2 matrix A as $f(x) = Ax$.

We assume that A is diagonalizable, that is, there exists an invertible matrix P such that $P^{-1}AP = \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}$ for some $\lambda, \mu \in \mathbb{R}$.

Note that $f^n(x) = (P^{-1}AP) \cdots (P^{-1}AP)x = P^{-1}A^nPx$.

f is hyperbolic at the origin $\iff |\lambda| \neq 1$ and $|\mu| \neq 1$.

There are three cases:

source $|\lambda|$ and $|\mu|$ are both larger than 1.

sink $|\lambda|$ and $|\mu|$ are both smaller than 1.

saddle One of $|\lambda|, |\mu|$ is larger than 1, the other is smaller than 1.

Hartman–Grobman Theorem

Theorem

Let p be a hyperbolic fixed point of $f : M \rightarrow M$. Then there exist neighborhoods $U_1, U_2 \subset M$ of p , and $V_1, V_2 \subset T_p M$ of $0 \in T_p M$, and a homeomorphism $h : U_1 \cup U_2 \rightarrow V_1 \cup V_2$ such that we have

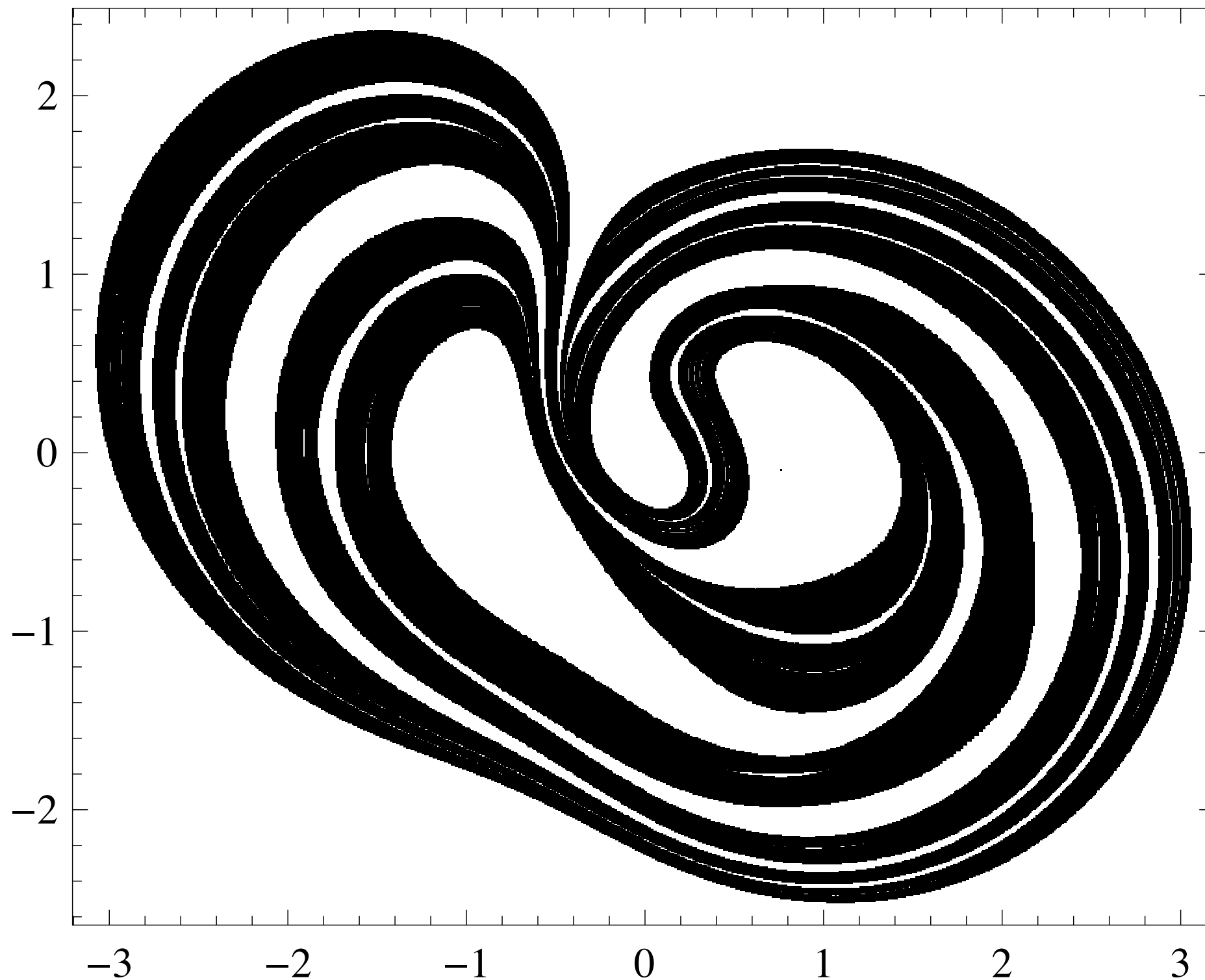
$$h(f(x)) = D_p f(h(x)) \quad \forall x \in U_1,$$

that is, the following diagram commutes:

$$\begin{array}{ccc} U_1 & \xrightarrow{f} & U_2 \\ h \downarrow & & \downarrow h \\ V_1 & \xrightarrow{D_p f} & V_2 \end{array}$$

This theorem implies that around a hyperbolic fixed point p , we can locally identify f with its differential $T_p f$.

Chaotic Invariant Set



Hyperbolic Periodic Points

Consider a diffeomorphism $f : M \rightarrow M$ on a manifold M .

At a point $x \in M$, f induces a linear map (the differential of f)
 $D_x f : T_x M \rightarrow T_x M$.

Definition

Let $p \in M$ be a fixed point of f , that is, $f(p) = p$. We say that p is **hyperbolic** if

$$D_p f : T_p M \rightarrow T_p M$$

has no eigenvalue with absolute value 1.

If p is a periodic point of f of period n , that is, $f^n(p) = p$, then we say that p is **hyperbolic** if it is a hyperbolic fixed point of f^n .

Uniform Hyperbolicity

Let $f : M \rightarrow M$ be a diffeomorphism and Λ a compact invariant set of f . Denote $T\Lambda := TM|_{\Lambda}$.

Definition

We say Λ is uniformly hyperbolic if $T\Lambda$ splits into a direct sum $T\Lambda = E^s \oplus E^u$ of invariant subbundles, and there exists constants $C > 0$ and $0 < \lambda < 1$ such that

$$\|Tf^n|_{E^s}\| < C\lambda^n \quad \text{and} \quad \|Tf^{-n}|_{E^u}\| < C\lambda^n$$

holds for all $n \geq 0$. Here $\|\cdot\|$ denotes a metric on M .

Hyperbolicity is Nice, but

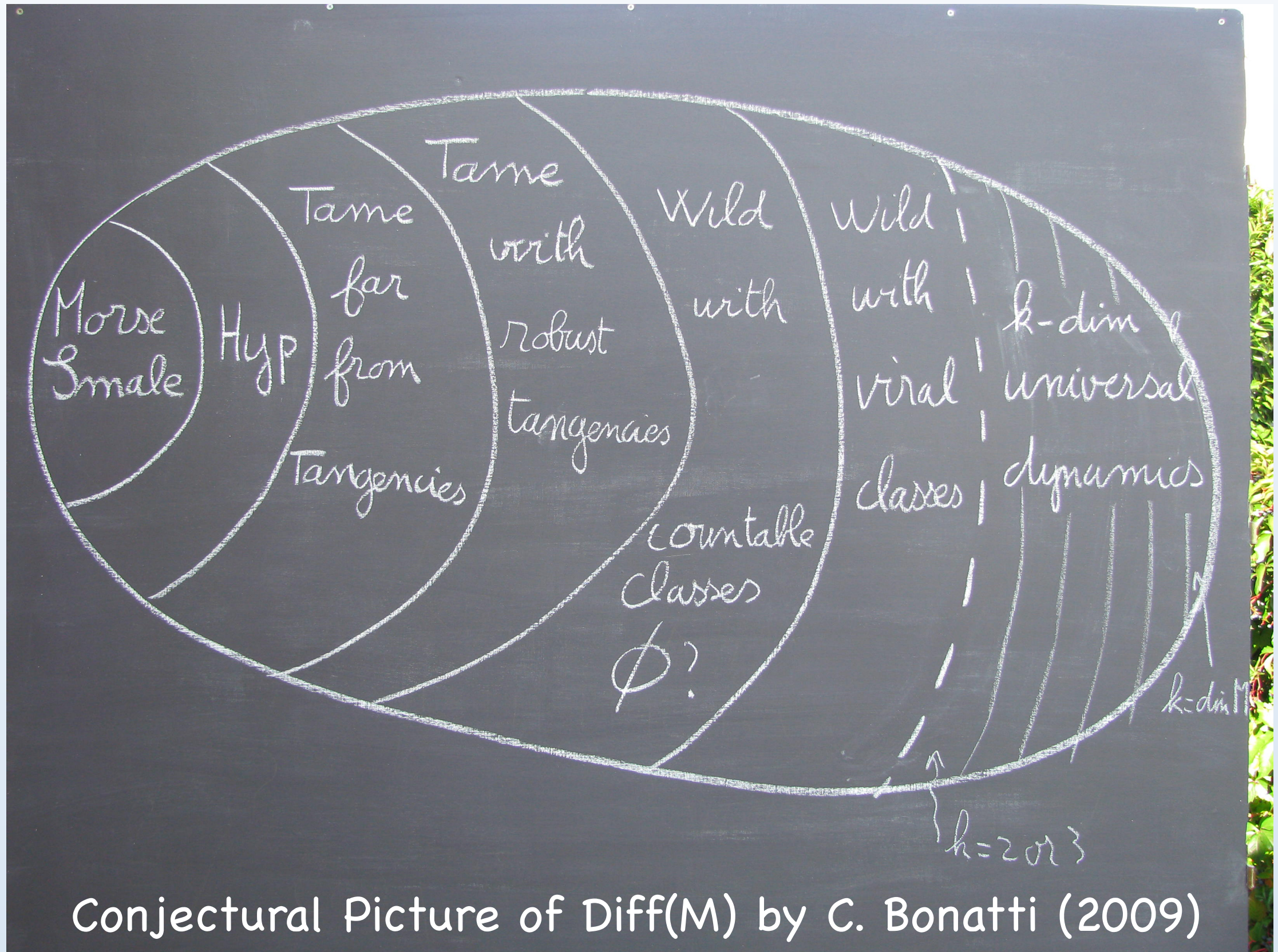
Un



Th
“un

In
sys

Ho
a g



olic

if
not.

Hyperbolicity of the Hénon

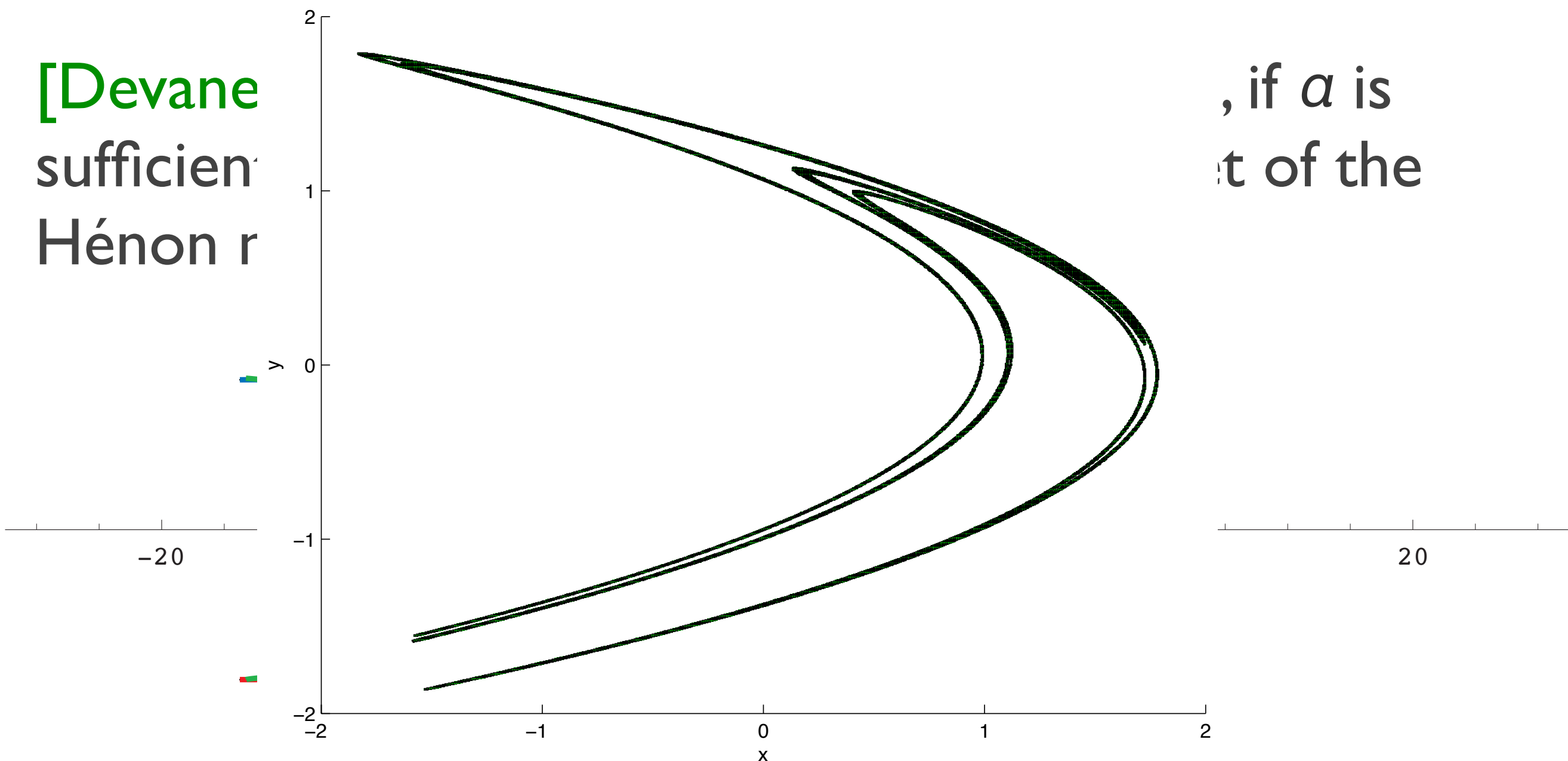
Consider the real Hénon map defined by

$$H_{a,b} : \mathbb{R}^2 \rightarrow \mathbb{R}^2 : (x, y) \mapsto (a - x^2 + by, x)$$

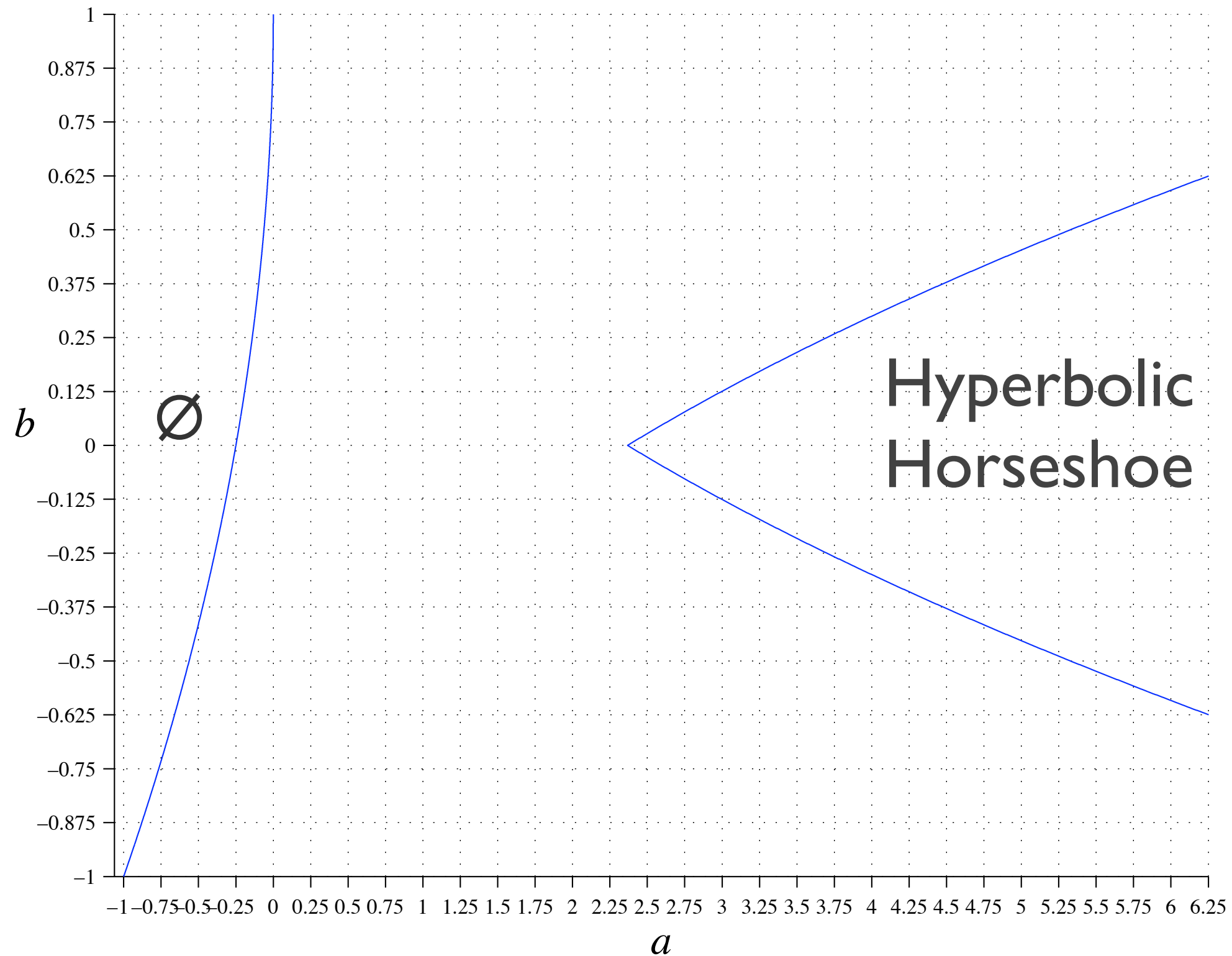
where a, b are real parameters.

[Devane
sufficien
Hénon r

, if a is
t of the

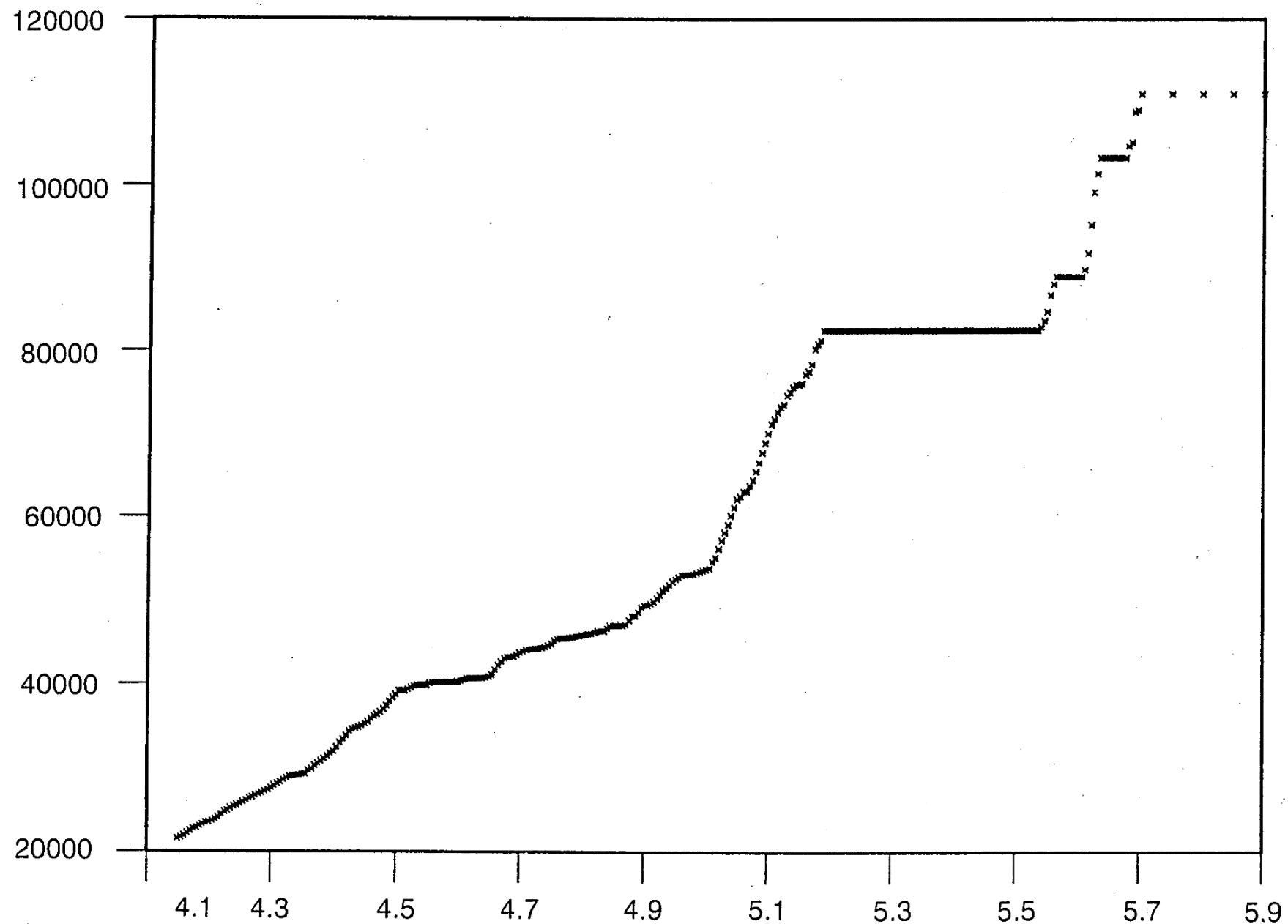


DN parameter region



Hyperbolicity of real Hénon

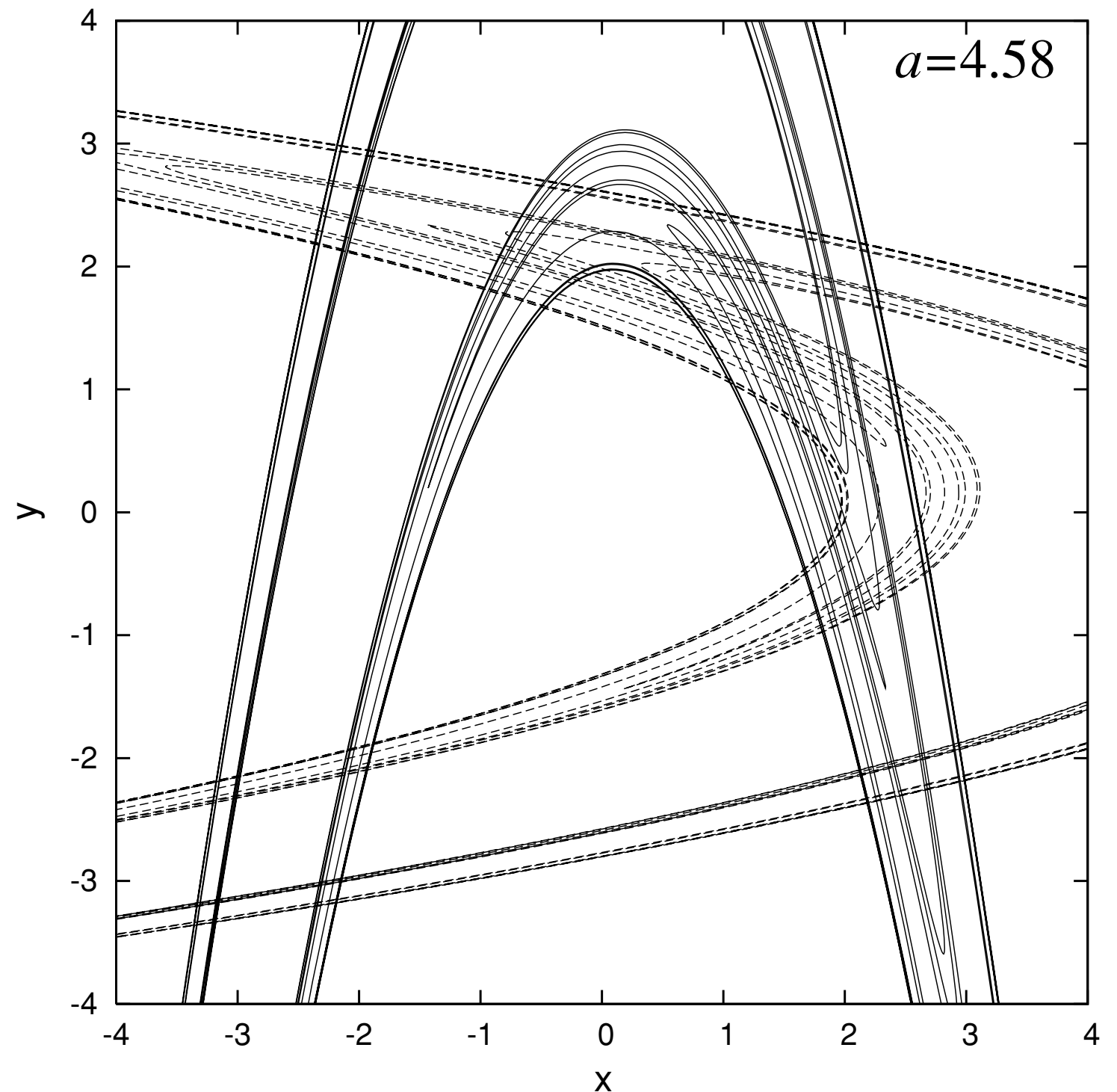
[Davis-MacKay-Sannami 1991] Conjectured the uniform hyperbolicity of the real Hénon map for $b = -1$ and $a = 5.4, 5.58, \dots$



Getting closer to tangency

A numerical picture of the invariant manifolds of $H_{4.58,-1}$.

Since the map is close to a tangency, it is very hard to construct and verify the hyperbolic splitting by hand.



Picture from Hagiwara-Shudo J. Phys A 34 (2004)

Computational Methods

Rigorous Algorithm for proving uniform hyperbolicity:

- * S. Hruska (2006)

Construct hyperbolic splittings using quadratic forms associated to the differential of the map.

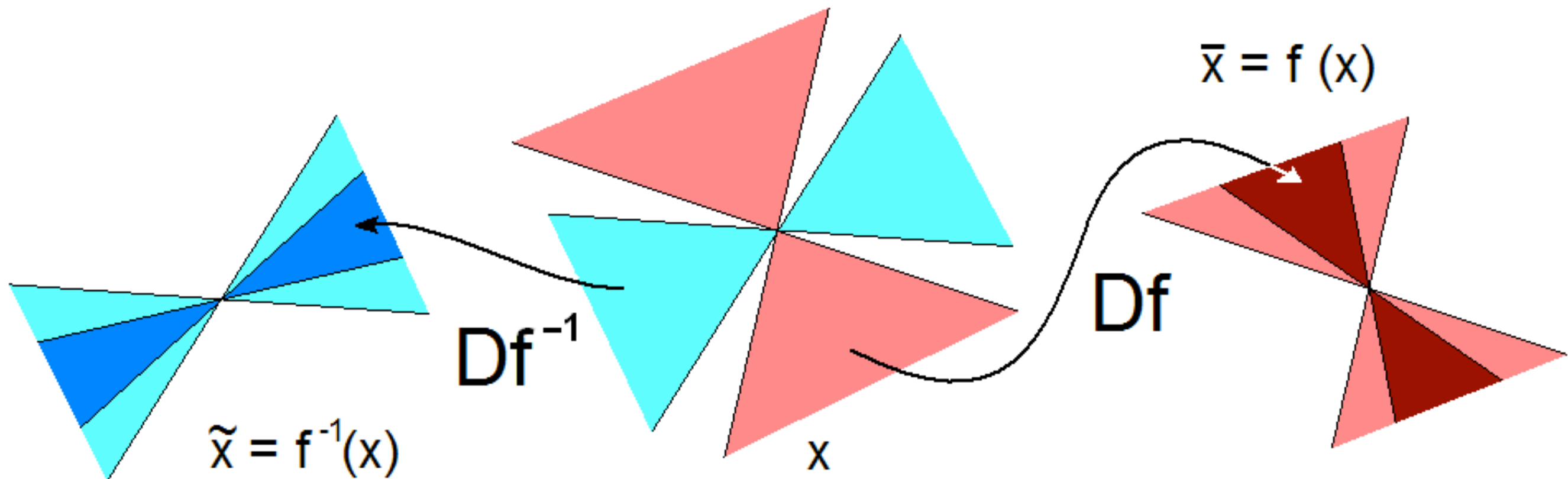
- * Z.A. (2007)

Use quasi-hyperbolicity to avoid the construction of hyperbolic splittings.

- * D. Wilczak (2010)

Close to Hruska's, but do not require the inverse of the differential. Best for hyp. attractors.

Stable and Unstable Cones



We say Λ is **box hyperbolic** if for each j such that $f(B_k) \cap B_j \neq \emptyset$,

If $v \in C_k^u$, $Df(v) \in C_j^u$ and $Q_j(Df(v)) > Q_k(v)$

If $v \in C_k^s$, $Df^{-1}(v) \in C_j^s$ and $Q_k(v) > Q_j(Df^{-1}(v))$.

Box Hyperbolicity

Theorem

Box hyperbolicity $\Rightarrow$ Uniform hyperbolicity

Proposition

Suppose for each B_k we have a nondegenerate Hermitian form $Q_k : \mathbb{R}^n \rightarrow \mathbb{R}$. Define the unstable and stable cones by

$$C_k^u := \{v \mid Q_k(v) \geq 0\}, \quad C_k^s := \{v \mid Q_k(v) < 0\}.$$

Then Λ is box hyperbolic if for each j such that $f(B_k) \cap B_j \neq \emptyset$,

$$(Q_j \circ Df) - Q_k$$

is positive definite.

Box Hyperbolicity Algorithm

1. Construct a box covering of Λ .
2. Find periodic points in Λ , and for each box B_k containing a periodic point, define Q_k using the hyperbolic splitting associated to the periodic point.
3. Spreading the Hermitian form constructed in the previous step by the action of Df , build Q_k for all k .
4. Check if $(Q_j \circ Df) - Q_k$ is positive definite.

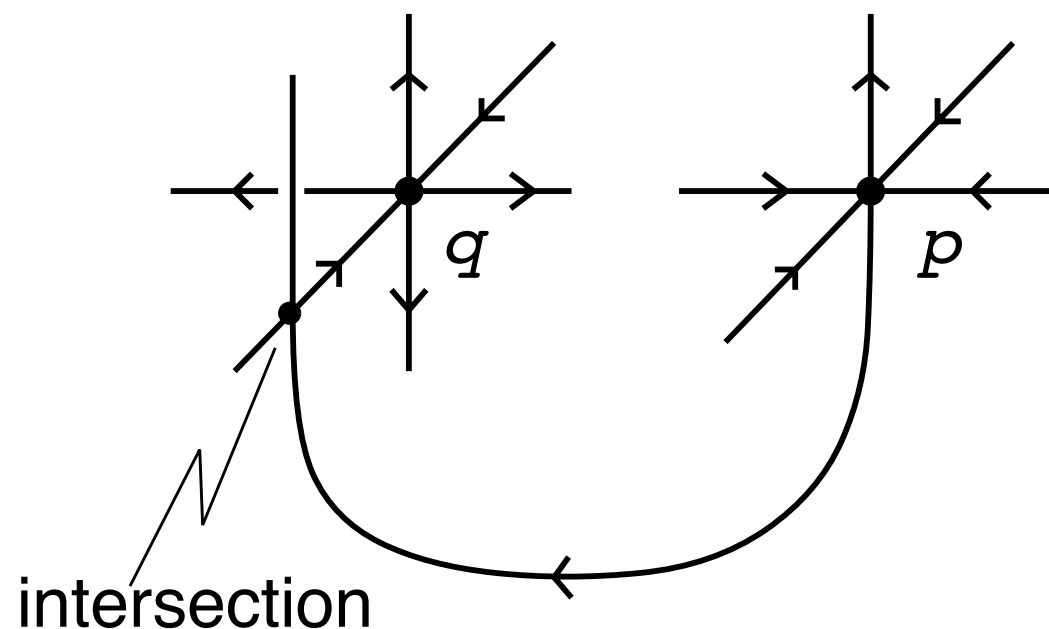
Quasi-Hyperbolicity

Definition. We say f is *quasi-hyperbolic* on Λ if $Tf : T\Lambda \rightarrow T\Lambda$ has no non-trivial bounded orbit.

uniform hyperbolicity $\Rightarrow$ quasi-hyperbolicity

quasi-hyperbolicity $\Rightarrow$ hyperbolicity of periodic orbits
non-existence of tangencies

quasi-hyperbolicity $\not\Rightarrow$ uniform hyperbolicity



$\Lambda := \{p\} \cup \{q\} \cup (W^u(p) \cap W^s(q))$
is quasi-hyperbolic, but is not
uniformly hyperbolic.

Chain Recurrence and q-Hyp

Theorem (Churchill-Franke-Selgrade, Sacker-Sell).

Assume $f|_{\Lambda}$ is chain recurrent. That is, $\mathcal{R}(f|_{\Lambda}) = \Lambda$.

Then f is uniformly hyperbolic on Λ if and only if f is quasi-hyperbolic on it.

Remark. The chain recurrent set is chain recurrent.

That is, $\mathcal{R}(f|_{\mathcal{R}(f)}) = \mathcal{R}(f)$ always holds (C. Conley).

To prove the uniform hyperbolicity of the chain recurrent set, it suffices to show that it is quasi-hyperbolic.

Rigorous Enclosures

Periodic orbits:

Theorem. The set of periodic points of f is contained in the cubes corresponding to cycles in $G(f)$.

Note that this theorem holds regardless of the stability of the periodic orbits.

The maximal invariant set and the chain recurrent set:

$\text{Inv } G(f) := \{v \in G(f) \mid \exists \text{ bi-infinitely long path through } v\}$

$\text{Scc } G(f) := \{v \in G(f) \mid \exists \text{ cycle through } v\}$

Theorem. The maximal invariant set of f is contained in $|\text{Inv } G(f)|$. The chain recurrent set of f is contained in $|\text{Scc } G(f)|$.

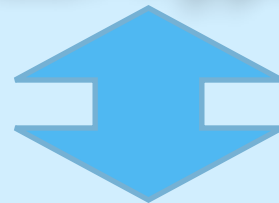
Isolating the Zero-Section

A compact set N is an *isolating neighborhood* if

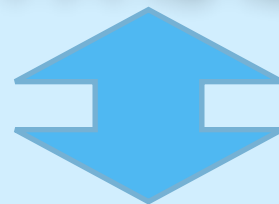
$\text{Inv}_f N := \{x \in N \mid f^n(x) \in N \text{ for all } n \in \mathbb{Z}\} \subset \text{int } N$.

An invariant set S is called an *isolated invariant set* if there is an isolating nbd N such that $\text{Inv}_f N = S$.

Λ is quasi-hyperbolic.



the 0-section of $T\Lambda$ is isolated w.r.t Tf .



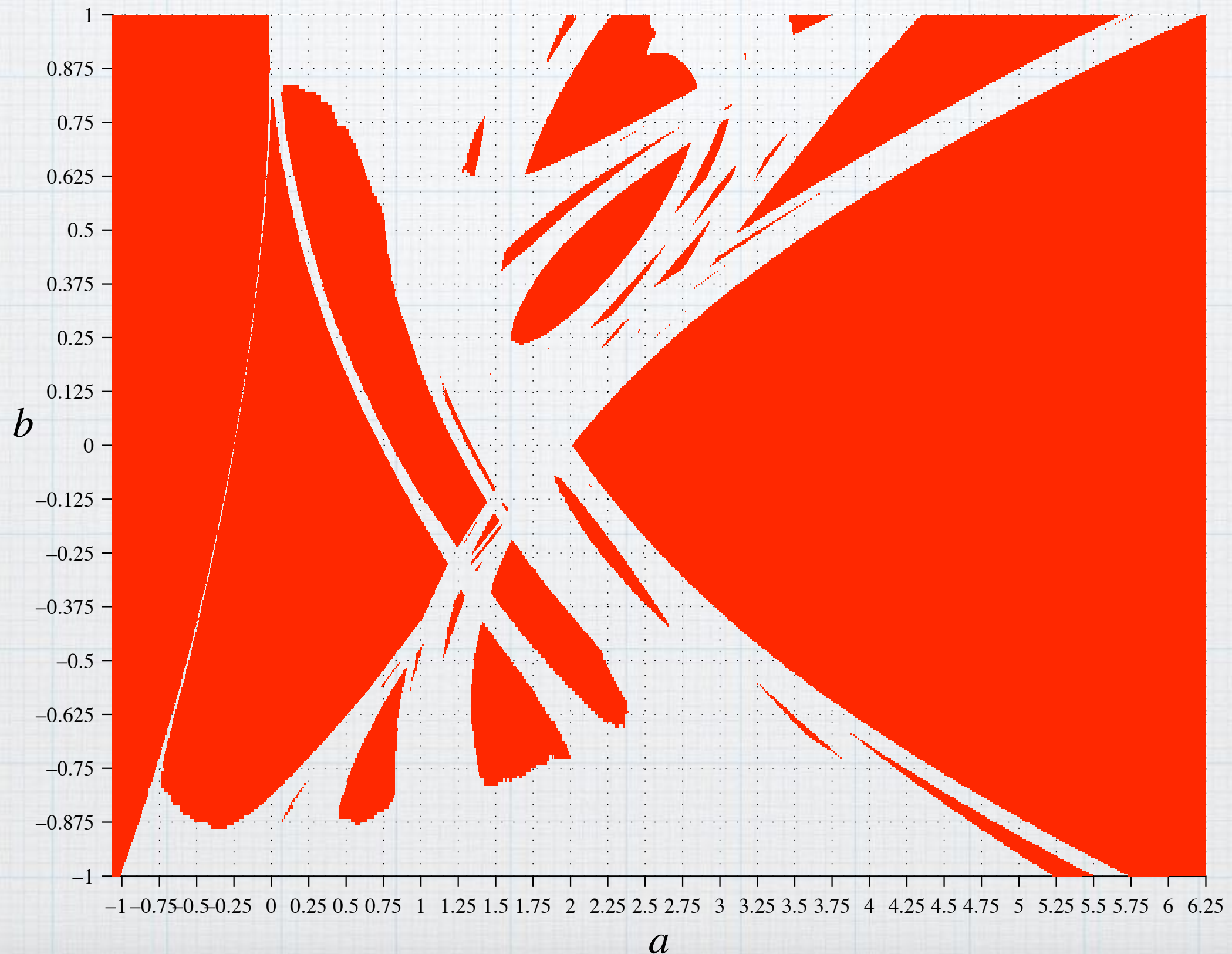
There exists an isolating neighbourhood that contains the 0-section of $T\Lambda$.

Hyperbolicity Algorithm

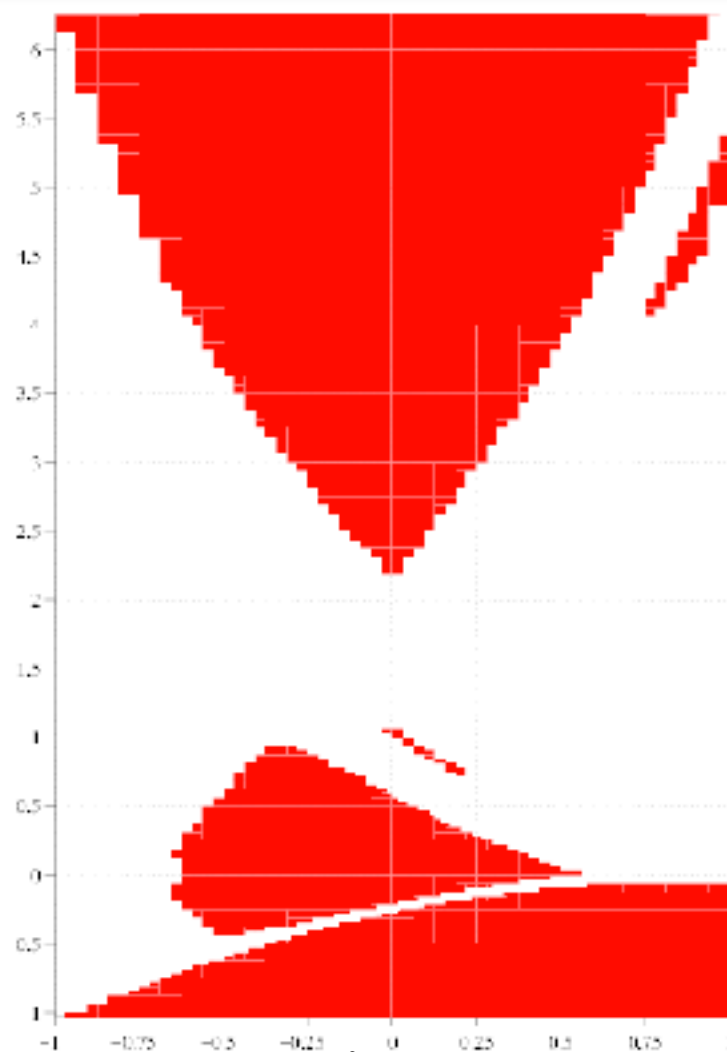
1. Find K such that $\mathcal{R}(f) \subset \text{int } K$. Let $N := K \times [-1, 1]^n$.
2. Compute $|\text{Scc } G(f, K)|$ and replace K with it.
Replace N with $N \cap (K \times [-1, 1]^n)$.
3. Compute $|\text{Inv } G(Tf, N)|$ and replace N with it.
4. If $N \subset K \times \text{int } [-1, 1]^n$ then stop.
Else, subdivide the cubes in K, N and goto 2.

Theorem. *If the algorithm stops, then $\mathcal{R}(f)$ is uniformly hyperbolic.*

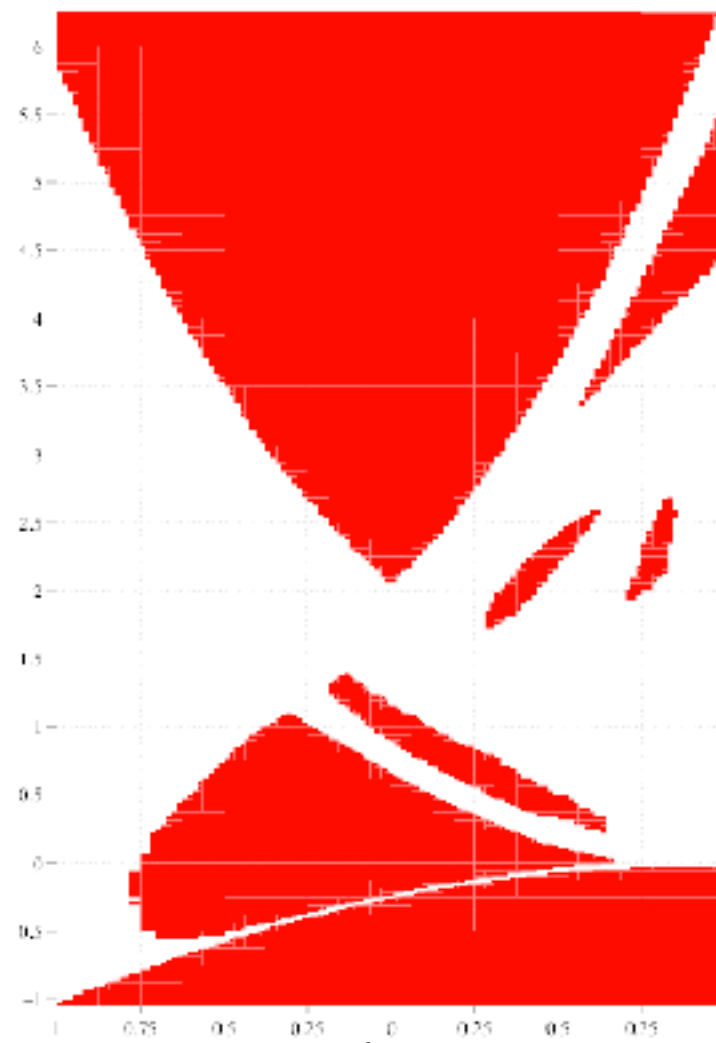
Theorem (ZA) If (a, b) is in the red region, then the chain recurrent set of $H_{a,b}$ is uniformly hyperbolicity.



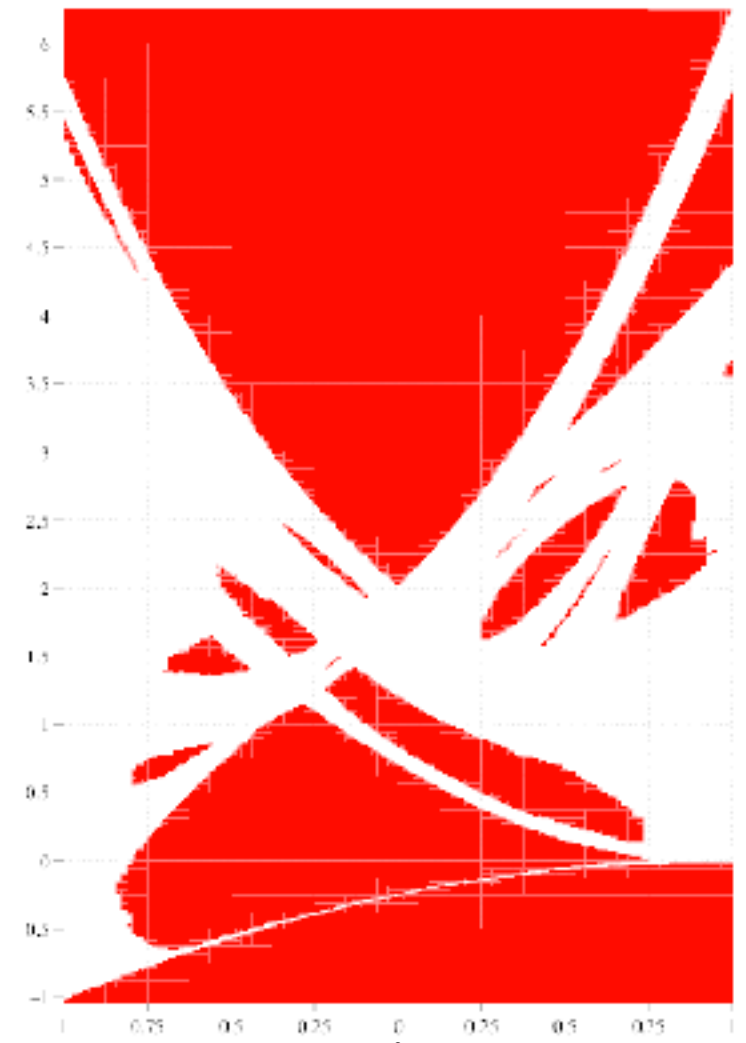
Convergence of Computation



1 hour



10 hours



100 hours

- Does this converge to the actual set of hyperbolic parameter values? Presumably yes.
- The hyperbolic parameters are not dense, due to Newhouse phenomena.
- How fast the convergence?

Computational Costs of Global Computations

Number of Vertices of $G(f)$

Suppose we have a grid decomposition $N = \cup N_i$
and want to compute a dynamical object S

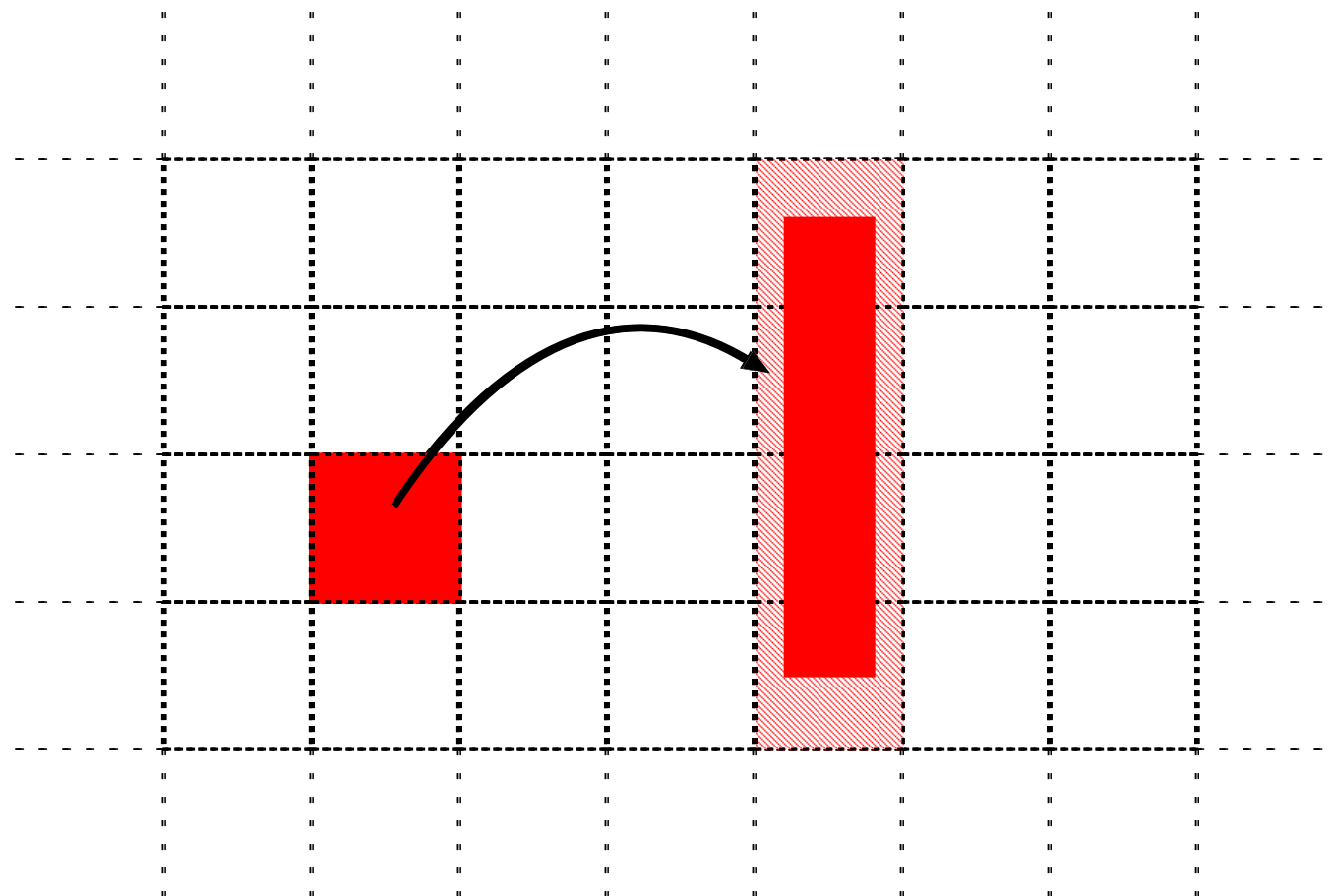
$$\begin{array}{ccccccc} S & \subset & \bigcup_{N_i \cap S \neq \emptyset} N_i & \subset & \begin{array}{c} |\text{Inv}G(f)| \\ |\text{Scc}G(f)| \\ \text{etc.} \end{array} & \subset & N \\ \text{dynamical} & & \text{optimal grid} & & \text{algorithmic} & & \text{entire} \\ \text{object} & & \text{covering} & & \text{outer approx.} & & \text{domain} \end{array}$$

- * The number of elements in the optimal grid covering can be estimated by the box dimension of S .
- * Difference between the outer approximation and the optimal grid covering is not computable in general (cf. non-computable Julia sets by Yampolski-Braverman) but for hyperbolic invariant sets, it can be estimated.

Number of Edges of $G(f)$

A rough estimate of the number of edges can be obtained from the Lyapunov exponent:

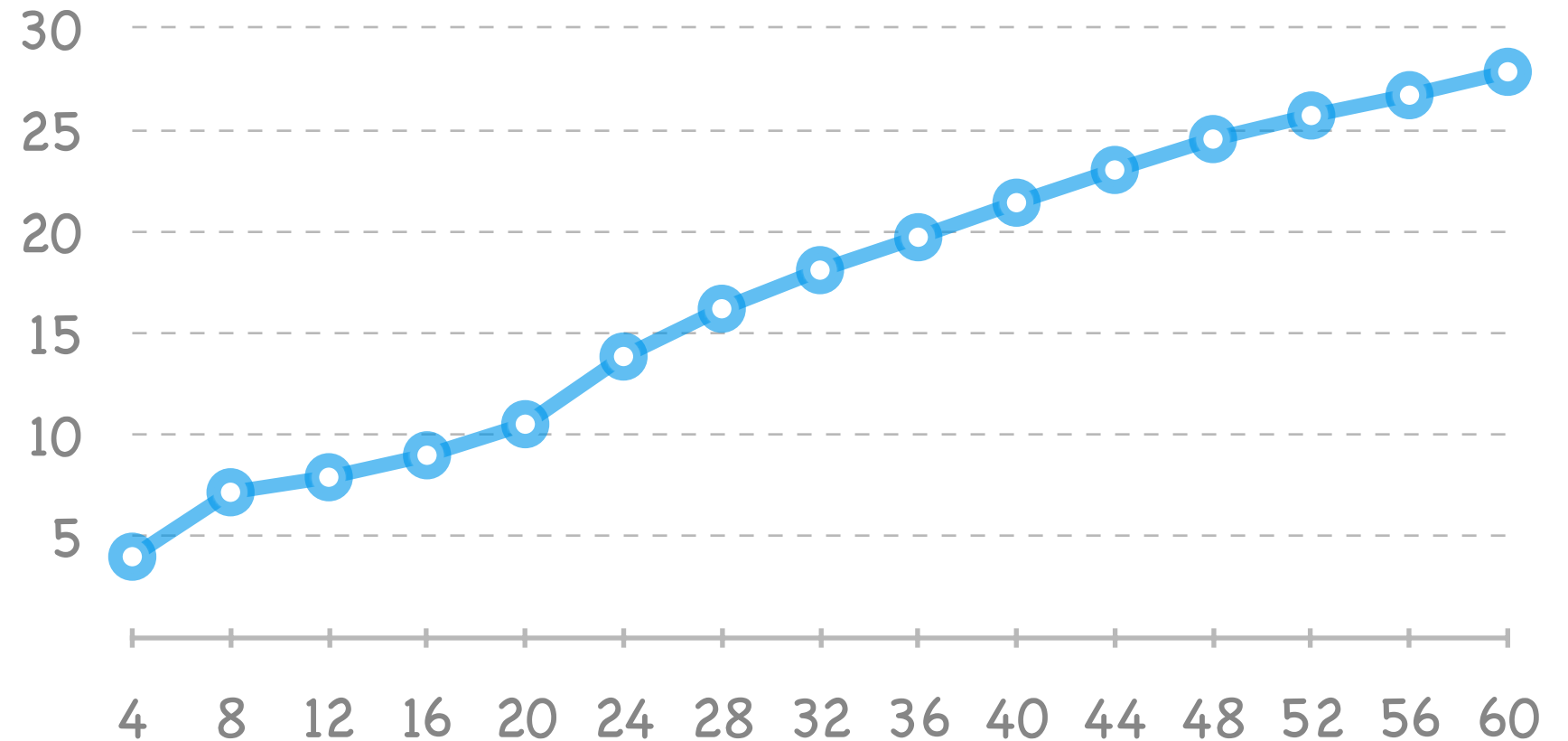
$$\text{average \#edges per vertex} \approx \prod_{\lambda_i \geq 0} \lceil \exp(\lambda_i) \rceil$$



Growth of the Data Size

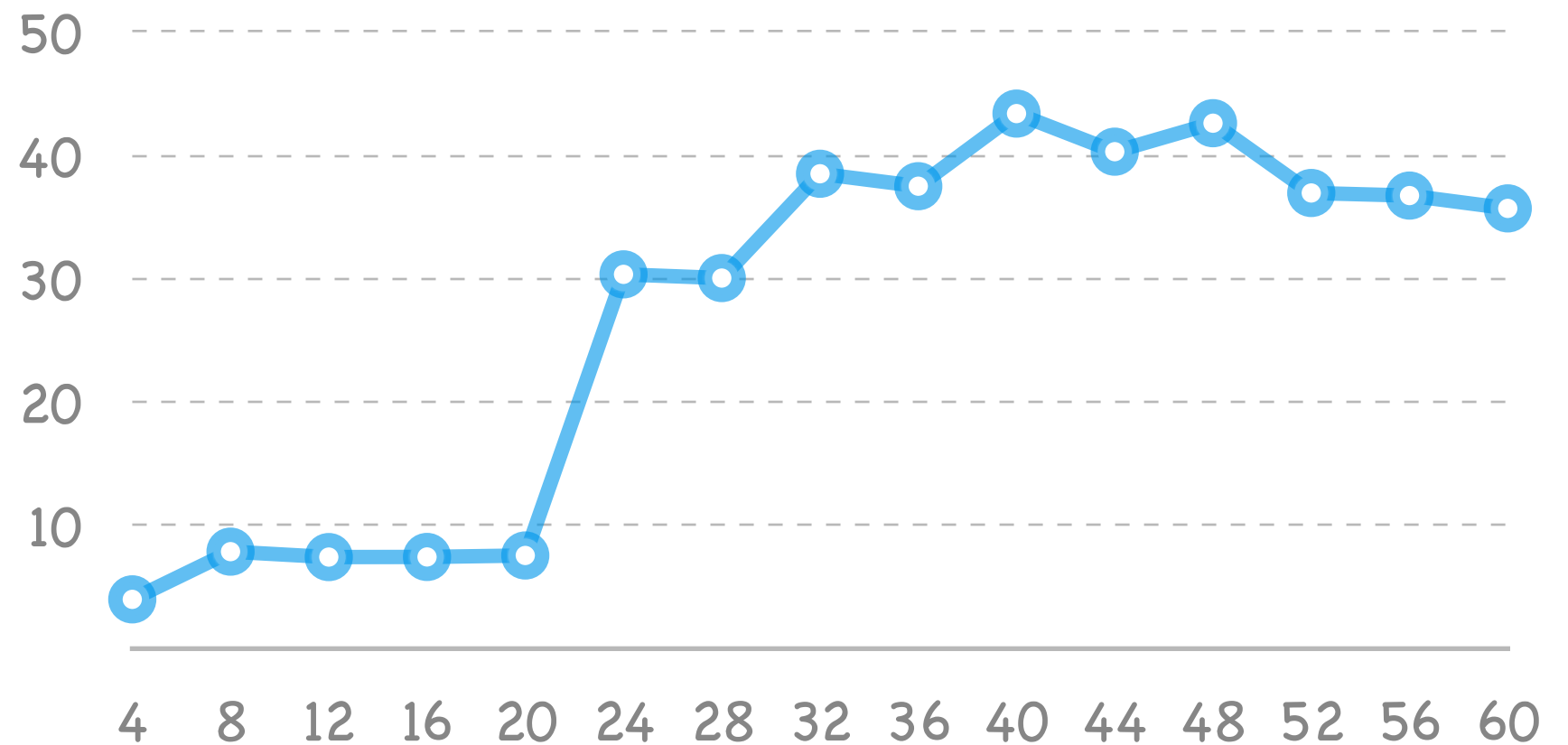
Vertices

subdivision
depth(horizontal)
vs
 $\log_2$ of #vertices



Edges

subdivision
depth(horizontal)
vs
the average
number of edges
per vertex



Inv Computation

Szymczak's algorithm to compute $\text{Inv}(G)$

```
do
   $n = \#G$  (the number of vertices)
  for each  $v \in G$ 
    if ( $v$  has no incoming edge) or
       ( $v$  has no outgoing edge)
      then put "remove" mark to  $v$ 
  end
  remove vertices with "remove" mark from  $G$ 
while ( $n \neq \#G$ )
```

* Typical # of do-while loop is less than 20

Inv Computation w/o Matrix

Szymczak's algorithm w/o Matrix

```
do
  n = #G (the number of vertices)
  for each v ∈ G
    compute outgoing edges from v and put
    "noremove" mark to their endpoints
    if (v has no outgoing edge)
      then put "remove" mark to v
  end
  remove vertices with "remove" w/o "noremove"
while (n ≠ #G)
```

SCC computation w/o Matrix

Tarjan's algorithm for Strongly connected comp.

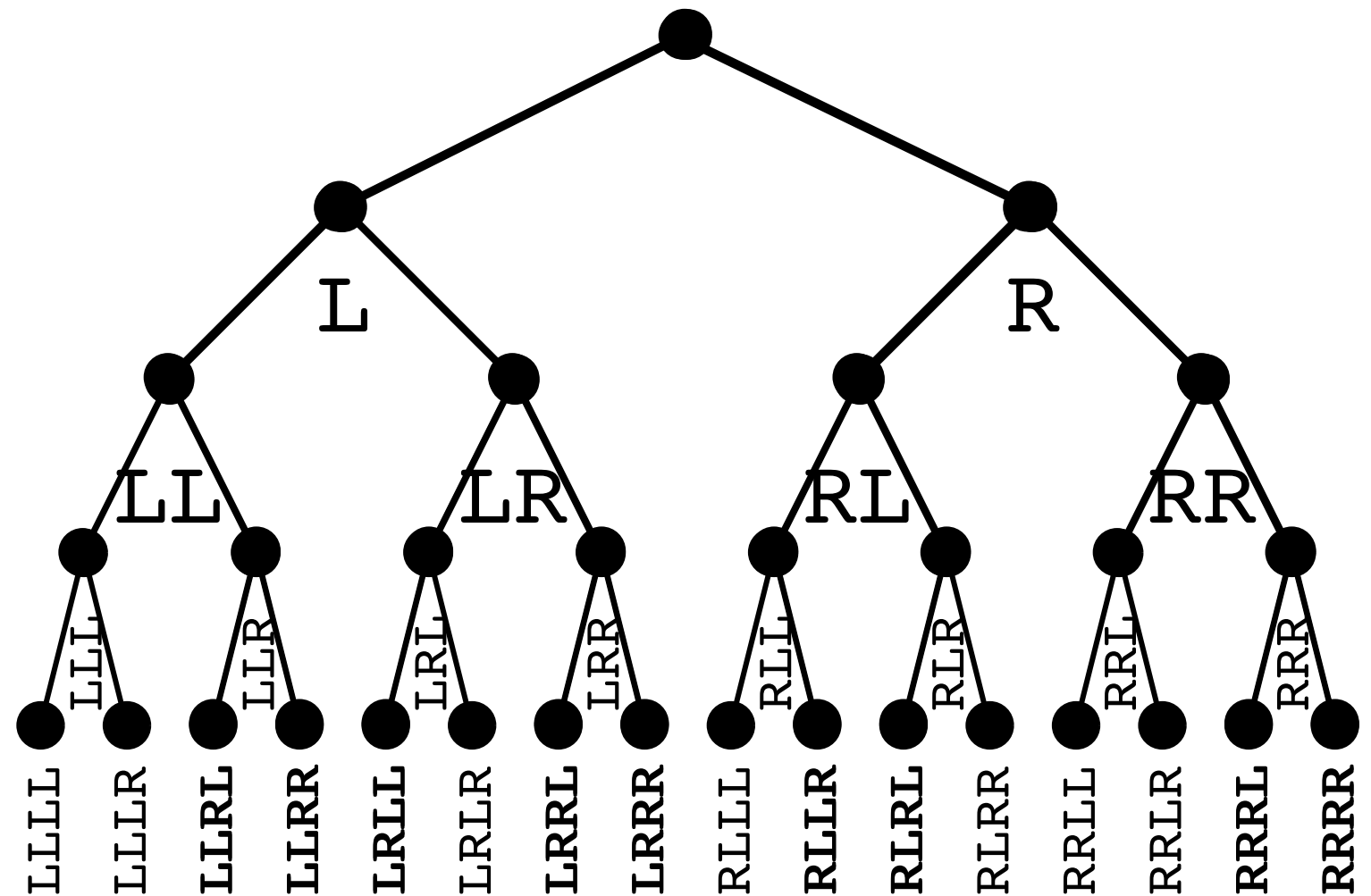
- * A variant of the depth-first search algorithm
- * Computational cost is linear ($\#V + \#E$)

Modified Tarjan's algorithm without matrix

- * Each time the depth-first iterator visits a vertex v , compute outgoing edges from v to vertices which has not visited yet
- * Computational cost for computing edges is
$$\log_2(\#vertices) \times \frac{n + 1}{2}$$
times larger (at worst) than the original
where n is the average $\#edges$ per vertices

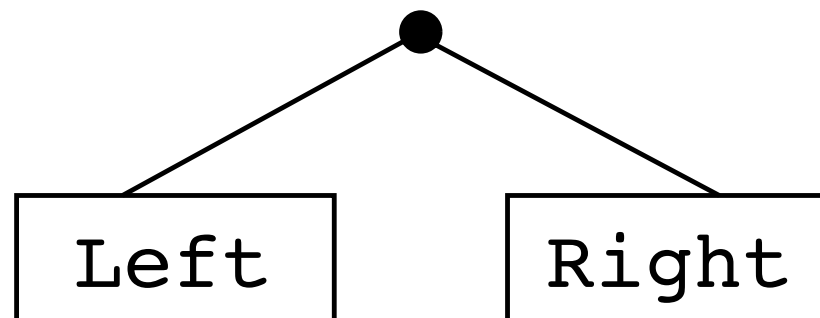
Binary Tree

In standard implementations such as GAIO, cubical spaces are represented by binary trees

[illegible]

Binary Tree by Pointers

Each node has two pointers: Left and Right



To access the left (right) child, simply read the value of the pointer Left (Right)

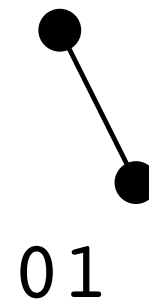
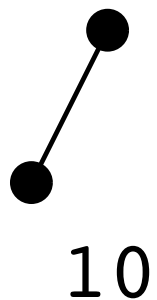
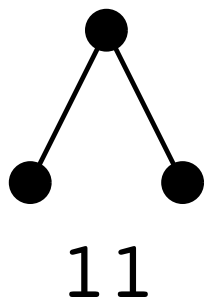
```
struct _Box {  
  
    byte sd;           /* coordinate in which box is/will be subdivided */  
    byte flags;        /* space for various flags */  
    byte color;        /* the color of that box */  
    Box *child[2];     /* pointer to the children */  
  
};
```

(From GAIO/include/gaio/Box.h)

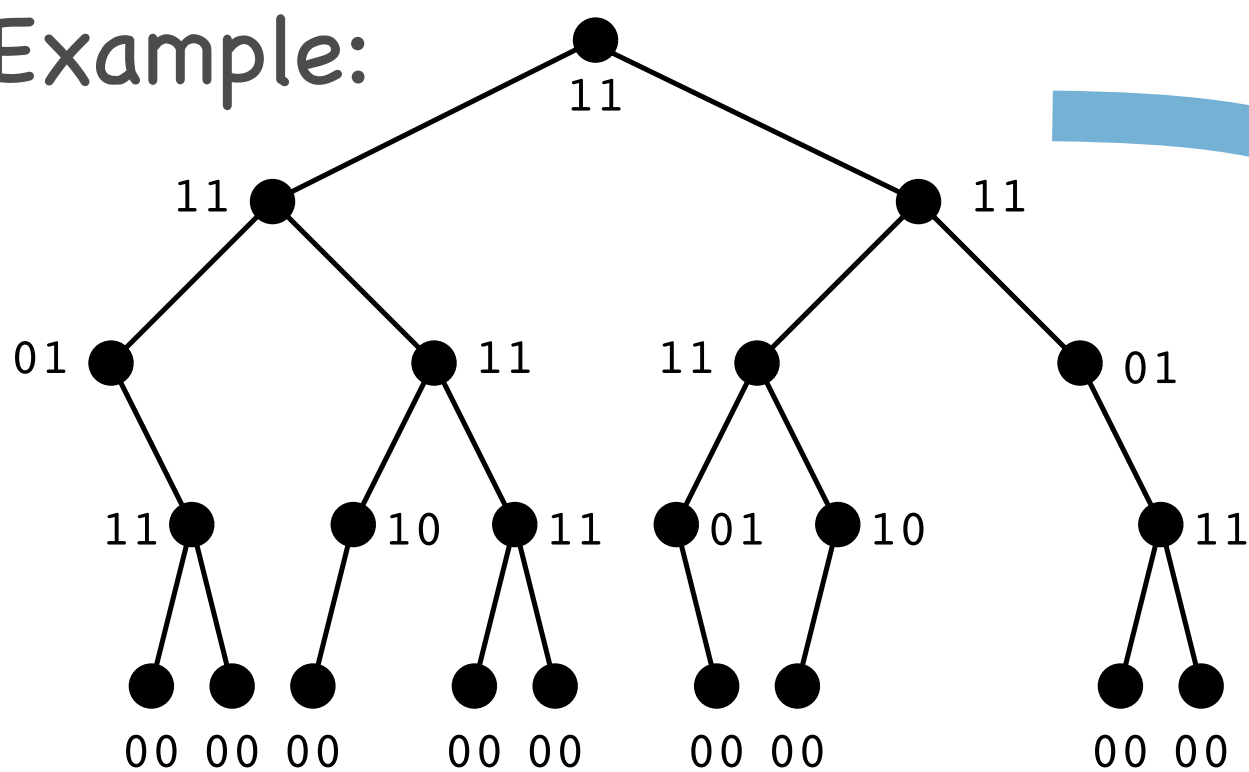
Problem: This requires at least 128 bits for one node.

Succinct Coding

Coding Rule:



Example:



```

      11
    11 11
  01 11 11 01
11 10 11 01 10 11
00 00 00 00 00 00 00 00 00
  
```

11111011111011110110110110000000000000000000

Rank/Select functions

$B[0, \dots, n-1]$: bit array of length n

$\text{rank}(B, i) :=$ the number of 1 in $B[0, i]$

$\text{select}(B, i) :=$ the position of i -th 1

For each node in the tree, assign an index $k \geq 0$ in the level ordering. Then we can show that:

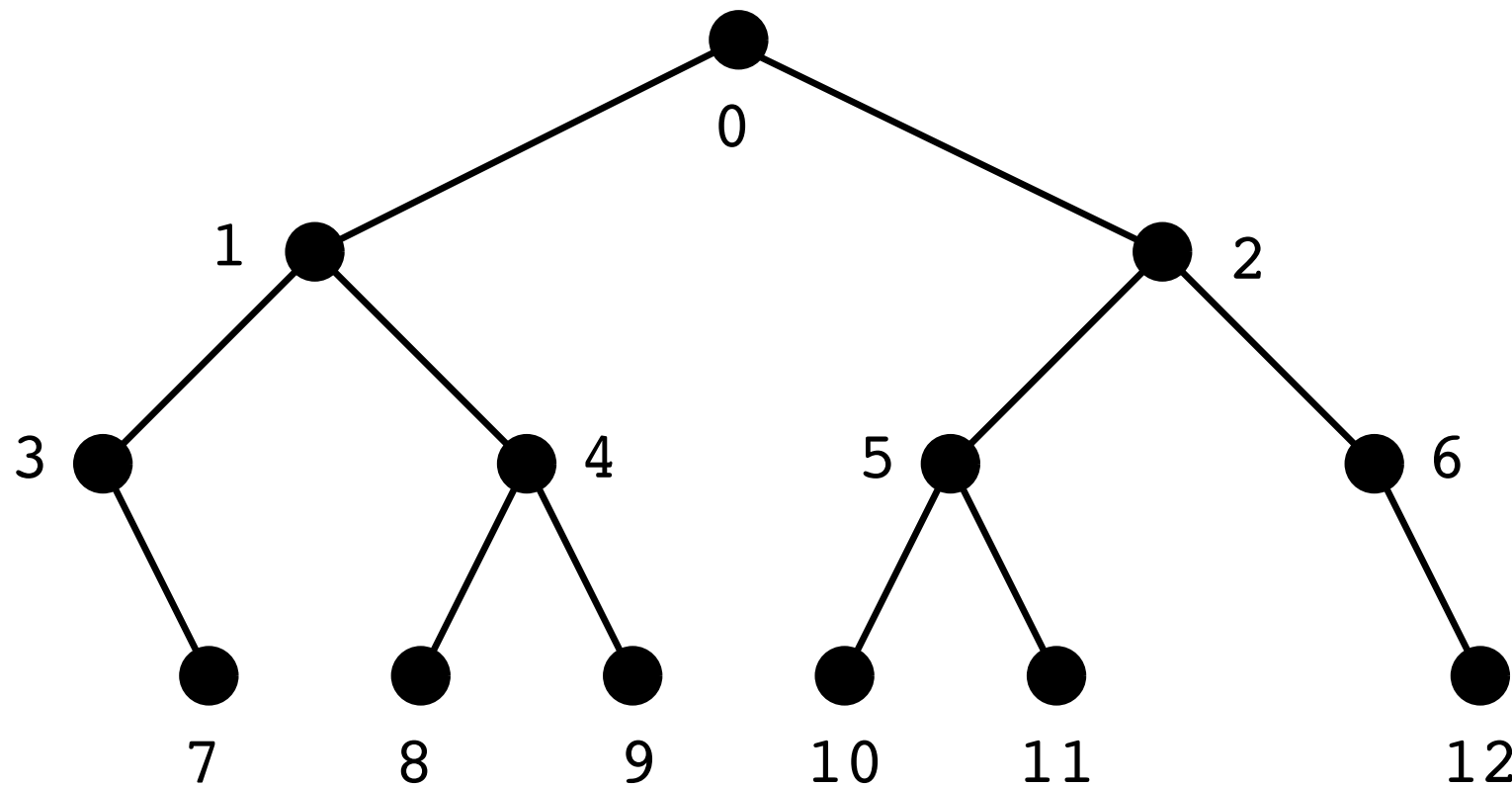
$$\text{left}(k) = \text{rank}(B, 2k)$$

$$\text{right}(k) = \text{rank}(B, 2k + 1)$$

$$\text{parent}(k) = \left\lfloor \frac{\text{select}(B, k)}{2} \right\rfloor$$

Rank/Select Example

$$\begin{aligned}\text{left}(k) &= \text{rank}(B, 2k) \\ \text{right}(k) &= \text{rank}(B, 2k + 1) \\ \text{parent}(k) &= \left\lfloor \frac{\text{select}(B, k)}{2} \right\rfloor\end{aligned}$$



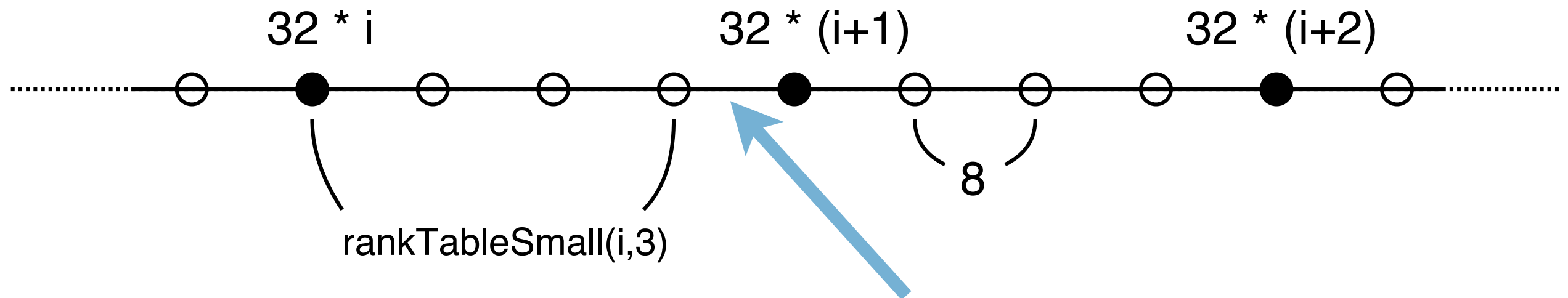
11	11	11	01	11	11	01	00	00	00	00	00	00
0 1	2 3	4 5	6 7	8 9	10 11	12 13	14 15	16 17	18 19	20 21	22 23	24 25

Efficient Rank Table

To compute $\text{rank}(k)$ efficiently, we prepare pre-computed tables.

unsigned long long int $\text{rankTableBig}(i) :=$
the number of 1 in the first $(32 * i)$ bytes

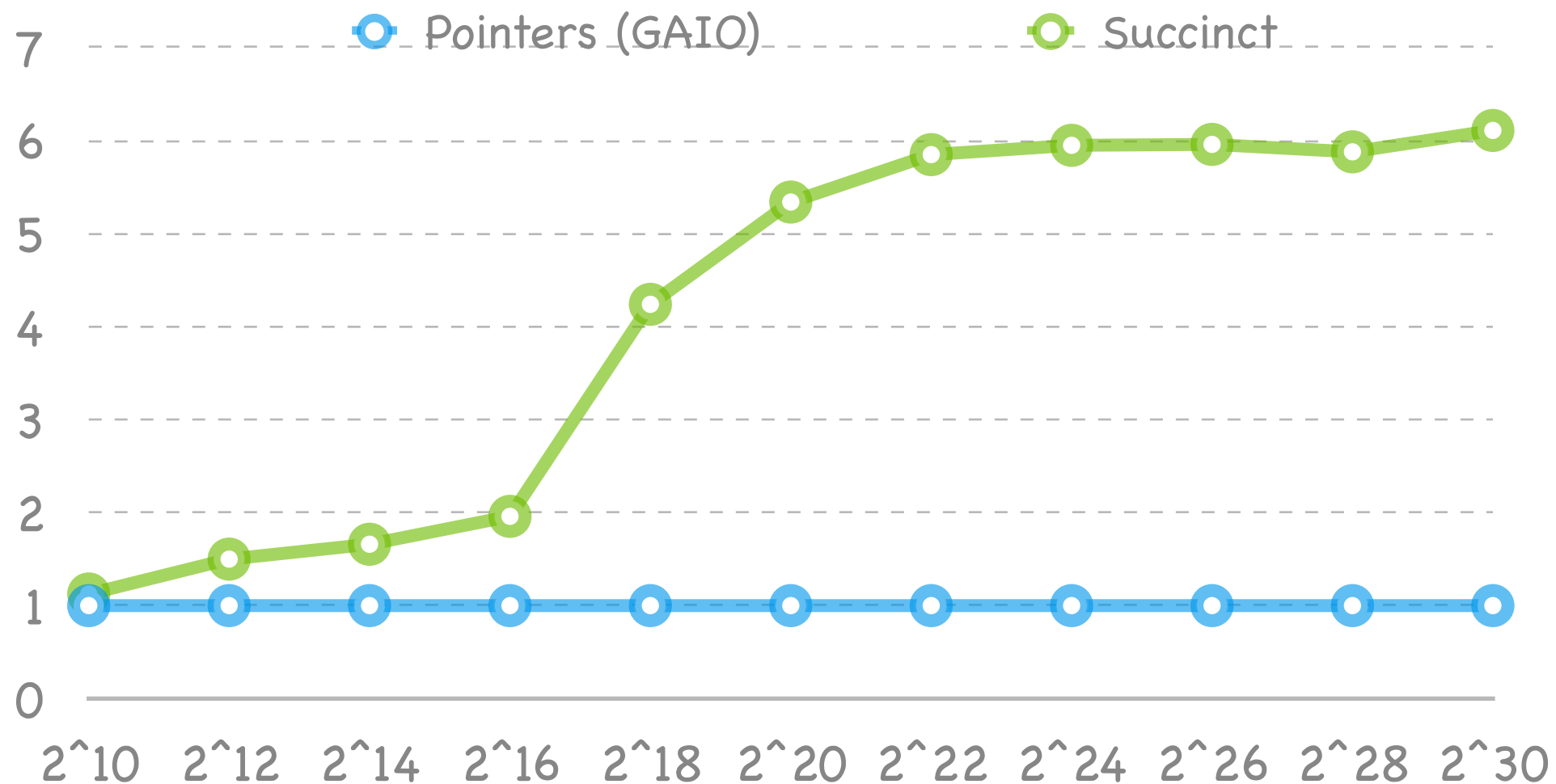
unsigned char $\text{rankTableSmall}(i, j) :=$ the number of 1 in
[($32 * i$)-th byte, ($32 * i + j * 8$)-th byte]



For example, if k -th bit is contained in this interval,
 $\text{rank}(k) = \text{rankTableBig}(i) + \text{rankTableSmall}(i, 3) + \text{the rest}$

Computational time is constant!

Growth of the data size



Comparison of CPU times for computing the max. inv. sets using Szymczak's algorithm.

Succinct data structure is about 6 times slower than GAIO.

Compressed Graph Computation

Binary Decision Diagram (BDD)

BDD is a data structure for efficiently storing boolean functions and can be used to represent a graph $G = (V, E)$:

- * Each vertex is identified with a binary number
- * A subset of V is a boolean function
- * Edges $E \subseteq V \times V$ is also a boolean function

SCC algorithms for BDD data structures

- * Xie-Beerel (ICCAD 1999)
- * Gentilini-Piazza-Policriti (Algorithmica 2008)

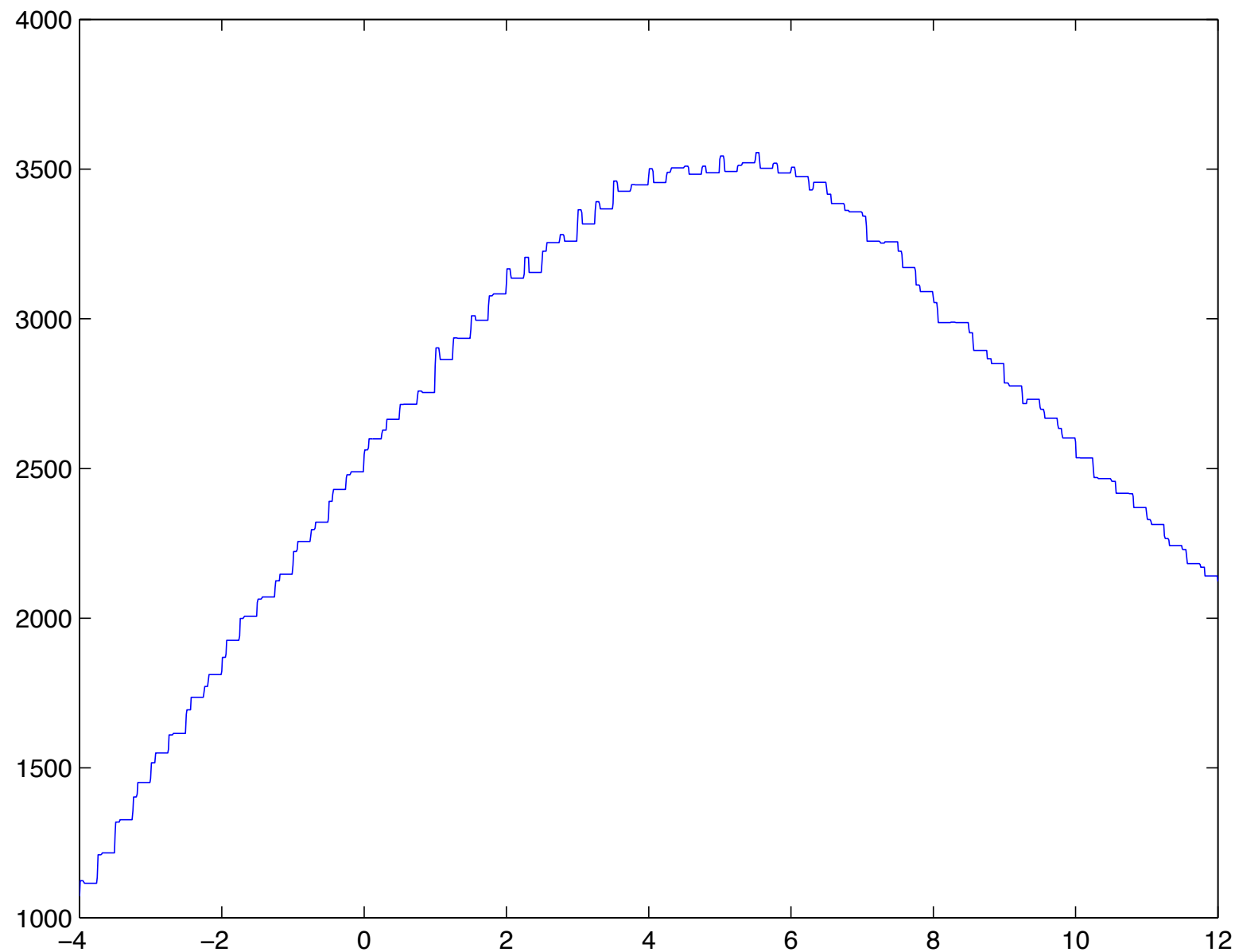
「フカシギの数え方」とZDD



ZDDアルゴリズム（湊ERATO@北大）の応用例

Information Theoretic Entropy vs Dynamical Entropy

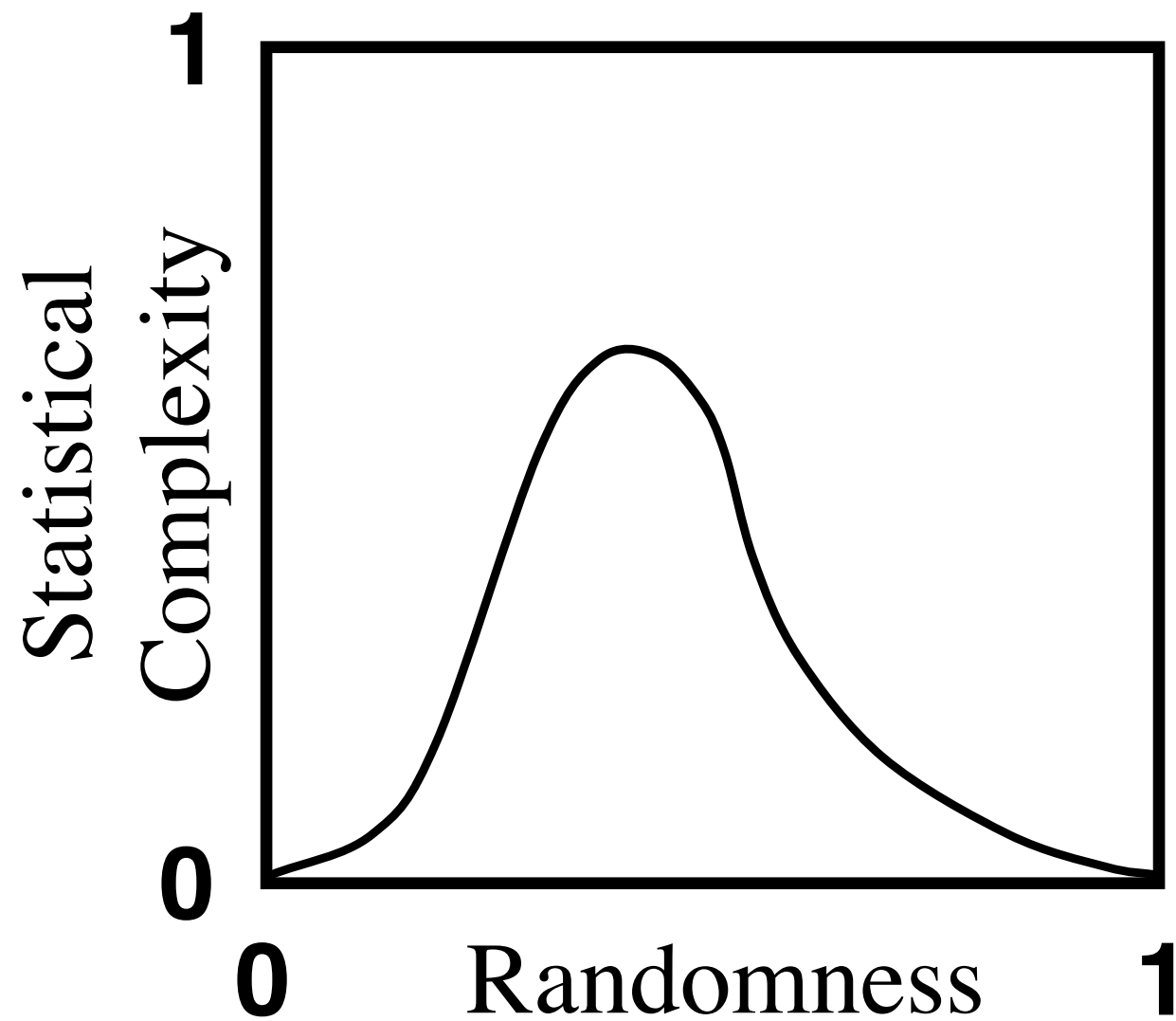
Computational Cost vs Entropy



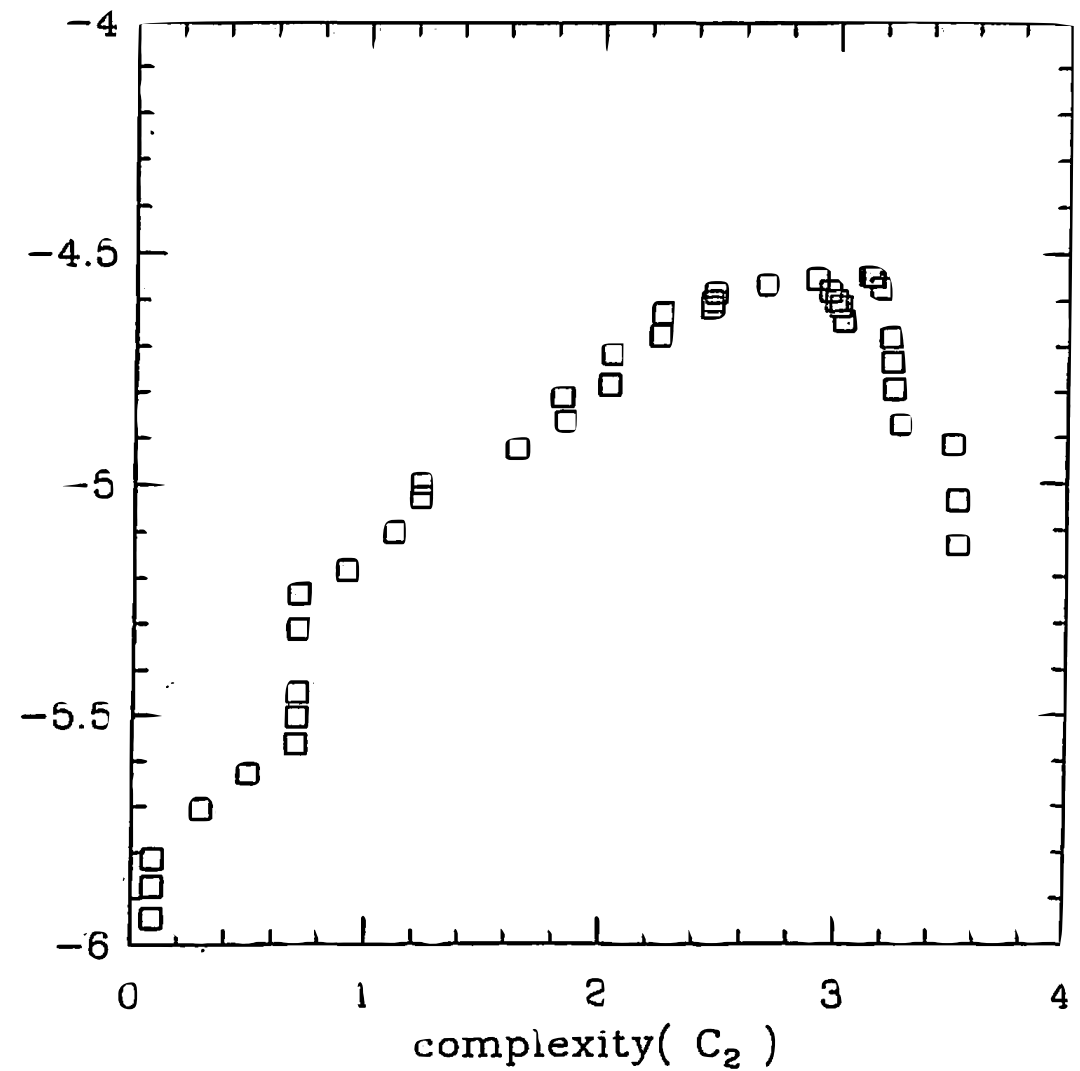
A parameter of the Hénon map vs compressed data size

- * The parameter roughly governs the nonlinearity
- * Computationally hard parameters are those with intermediate topological entropy

Randomness vs Complexity



J.P. Crutchfield, Physica D (1994)



Ikegami-Tsuda
Inter. J. Modern Physics C (1992)

Symbolic Dynamics

The shift space

$\{0, 1, \dots, n-1\}$: the alphabets

$\Sigma := \{0, 1, \dots, n-1\}^{\mathbb{Z}}$: the shift space

The metric on the shift space

For $s = (\dots s_{-2} s_{-1} \cdot s_0 s_1 \dots)$ and $t = (\dots t_{-2} t_{-1} \cdot t_0 t_1 \dots)$,

$$d(s, t) := \sum_{-\infty}^{\infty} \frac{1 - \delta(s_i, t_i)}{2^{|i|}} \quad (\delta \text{ is the Kronecker delta})$$

Two sequences are close if their entries in the middle agree

The shift map

$$\sigma : \Sigma \rightarrow \Sigma : (\dots s_{-2} s_{-1} \cdot s_0 s_1 \dots) \rightarrow (\dots s_{-1} s_0 \cdot s_1 s_2 \dots)$$

homeomorphism

Subshift of Finite Type

Transition Matrix

$A : n \times n$ matrix with elements in $\{0, 1\}$

Subshift Space

$$\Sigma_A := \{s \in \Sigma \mid A_{s_i s_{i+1}} = 1 \text{ for all } i \in \mathbb{Z}\} \subset \Sigma$$

The shift map $\sigma : \Sigma_A \rightarrow \Sigma_A$

Example:

$$n = 2$$

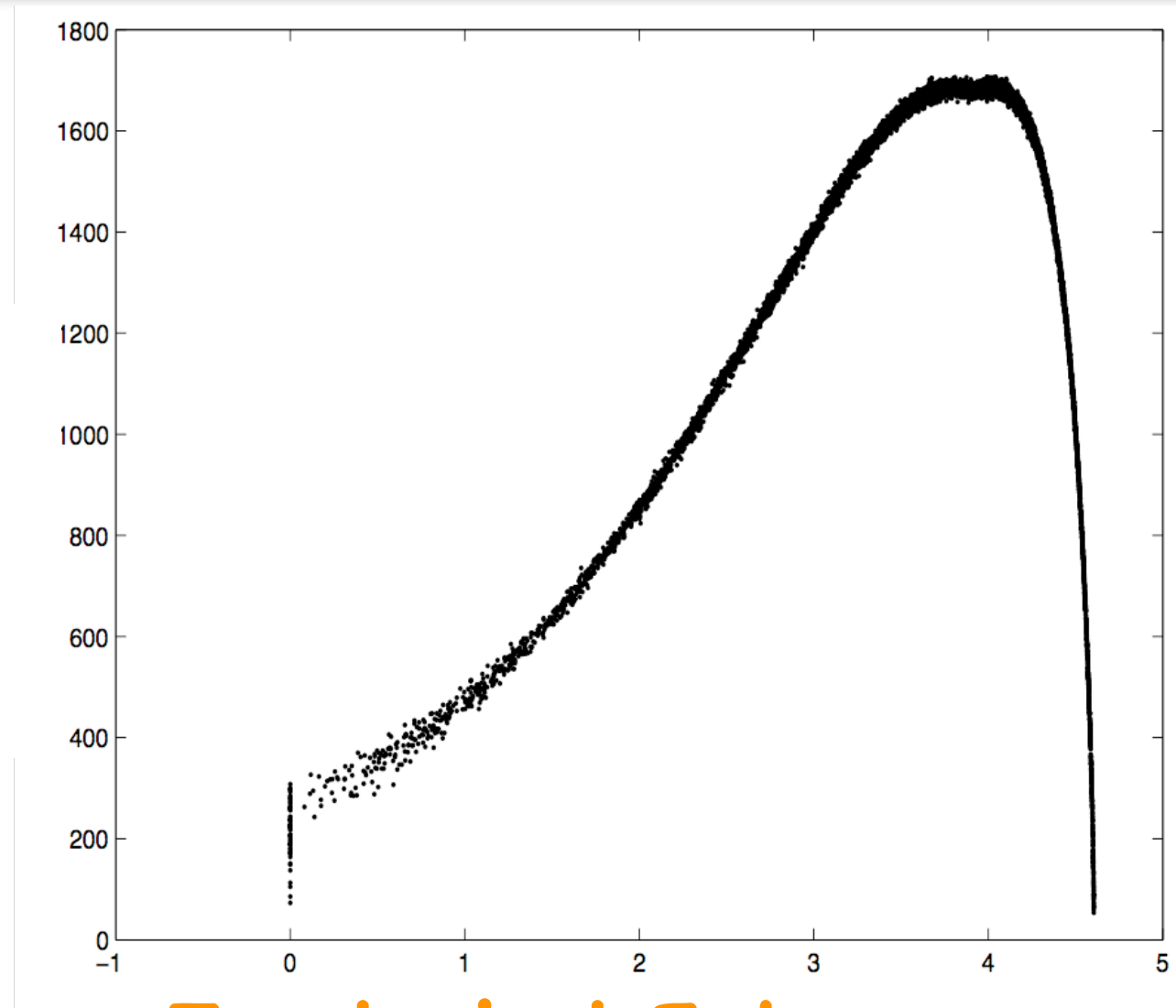
$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$



“0” can be followed by both “0” and “1”,
but “1” can only be followed by “0”

A simple experiment

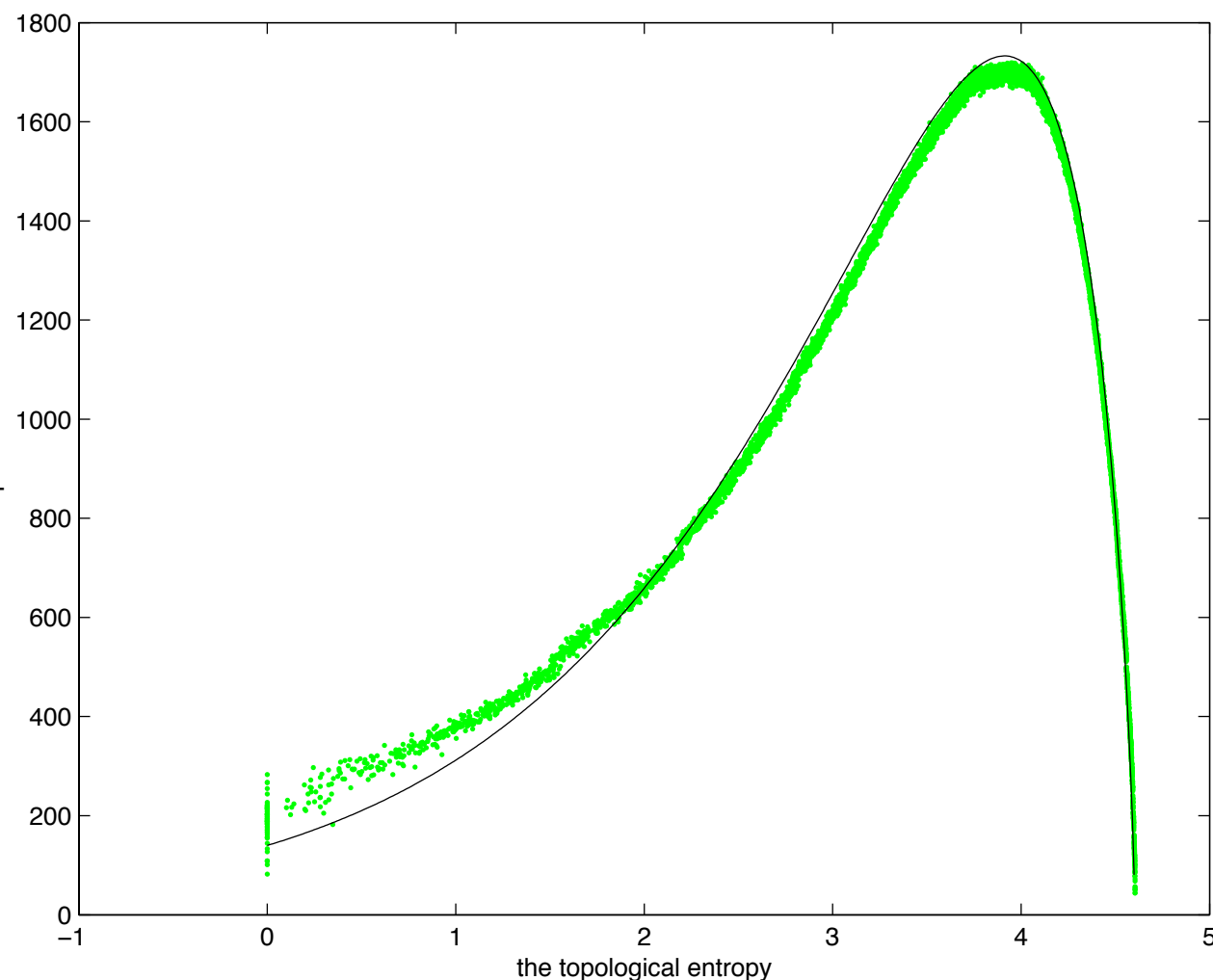
**bzip2 compressed
filesize of the matrix**



Topological Entropy -->

The topological entropy vs the compressed data size of matrix for 10,000 randomly generated 0-1 transition matrices of size 100×100 (density of 1 is equally distributed on $[0, 1]$).

Shannon/Topological Entropies



n : the size of the transition matrix

The data points accumulate on $y = -d \log d - (1 - d) \log(1 - d)$ where $d = \exp(x - \log n)$

- * bzip2 is an entropy coding
 $\Rightarrow$ data size $\approx$ Shannon entropy
- * Shannon entropy is a function of the density.
- * Topological entropy can be estimated by the density using Perron-Frobenius theorem.

Summary

- * In graph theoretical computations for dynamical systems, the size of matrices representing the graph can be very large.
- * The size of matrix depends the geometric and dynamical behaviors of the system
- * It seems that there exist interesting relations between the entropy of a dynamical systems and the amount of information required to define it.

Thank you!

Computability of Julia sets

Computability of the Julia sets

Theorem (Braverman-Yampolsky): For any $c \in \mathbb{C}$, the filled Julia set K_c is computable (by an oracle Turing machine). On the other hand, there exists a computable $c \in \mathbb{C}$ such that J_c is not computable (by any oracle Turing machine).

Ref: M. Braverman and M. Yampolsky, Computability of Julia sets, Springer-Verlag

What is Computability?

Computability of a real number: We say $x \in \mathbb{R}$ is computable if there exists a computer program which inputs $k \in \mathbb{N}$ and outputs in finite time a rational number $\phi(k)$ such that $|x - \phi(k)| < 2^{-k}$ holds.

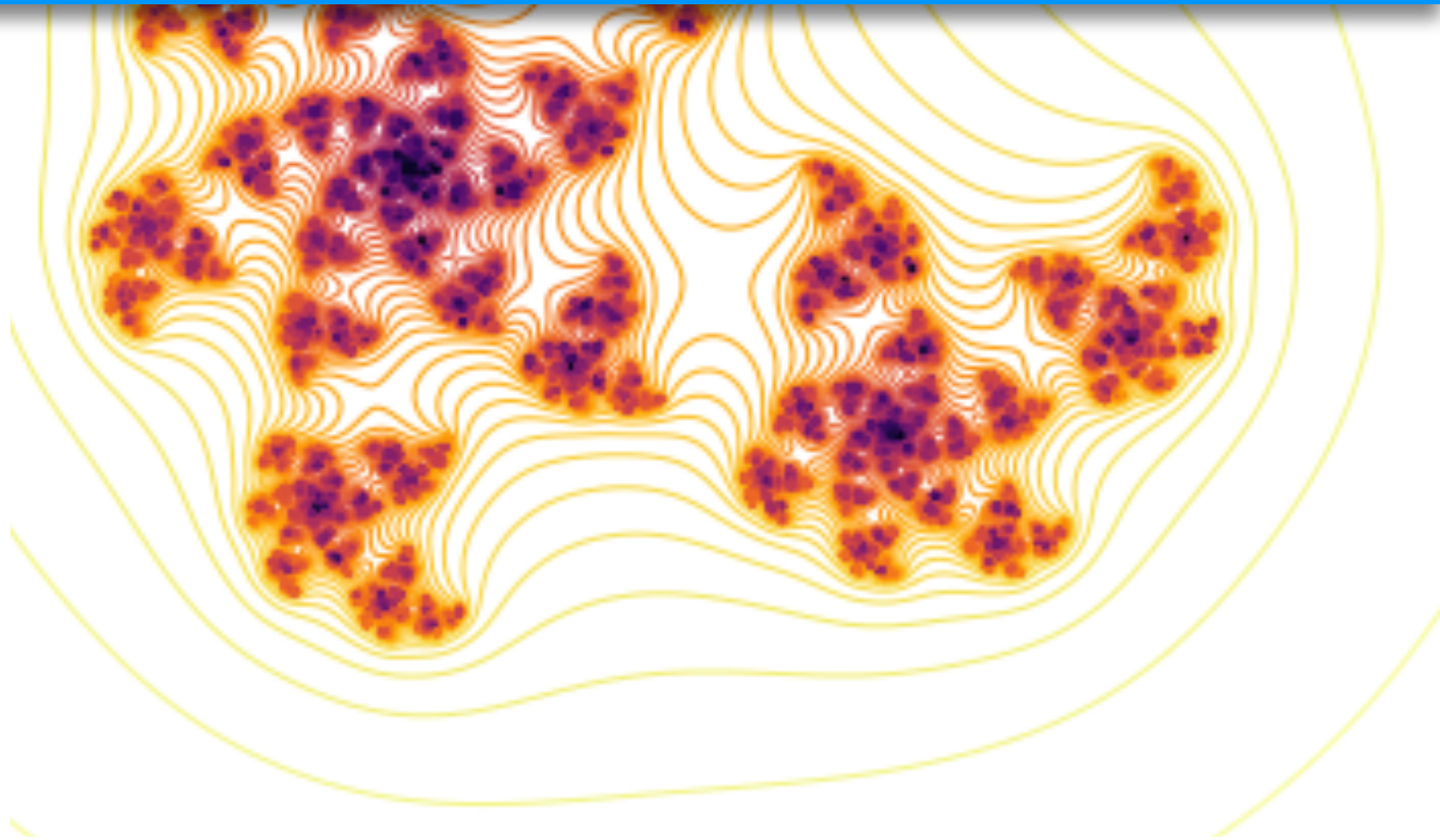
- ▶ Almost all real numbers are non-computable.
- ▶ $z \in \mathbb{C}$ is computable $\iff \operatorname{Re} z, \operatorname{Im} z$ are computable.

Computability of a set: We say $K \subset \mathbb{C}$ is computable if there exists a computer program which inputs $k \in \mathbb{N}$ and outputs in finite time a set $\phi(k) \subset \mathbb{C}$ such that its Hausdorff distance to K is less than 2^{-k} .

Computability of K_c

Outer approximation:

Using level curves of the Green function, we can construct an outer approximation of K_c effectively.



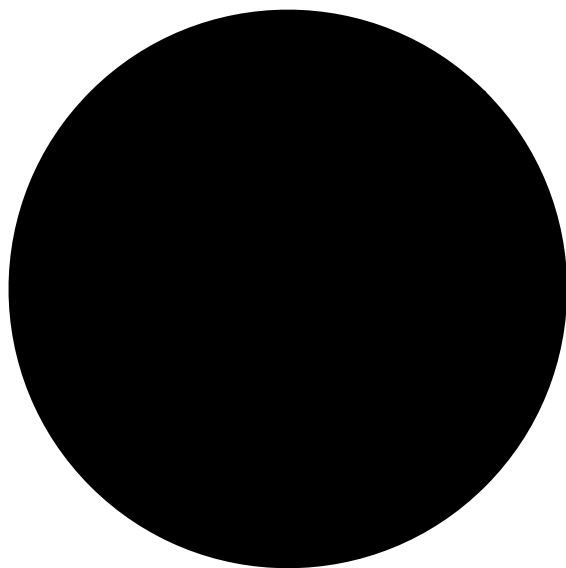
Inner approximation: Dynamics on each Fatou component (= components of $\mathbb{C} \setminus J_c$) is classified into three types (attractive basin, parabolic basin, Siegel discs) and for each type, we can construct an effective inner approximation.

Non-computability of J_c

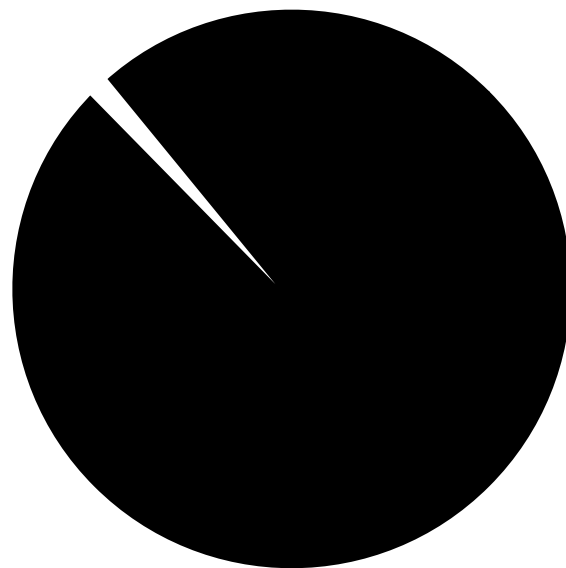
- $\exists$ non-computable Julia set which has a Siegel disc

Schematic picture:

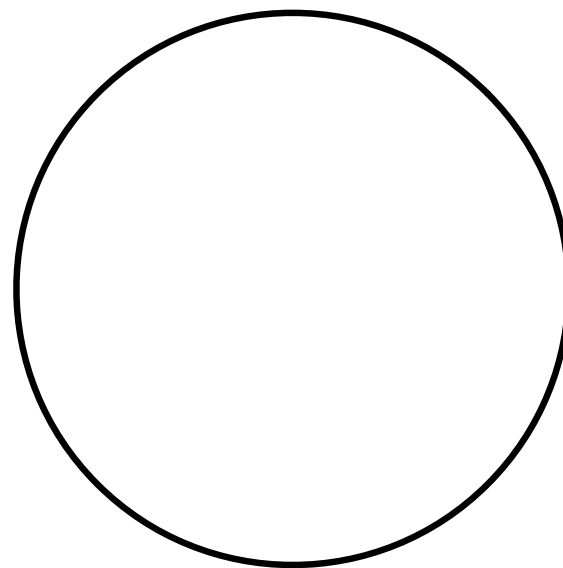
K_c



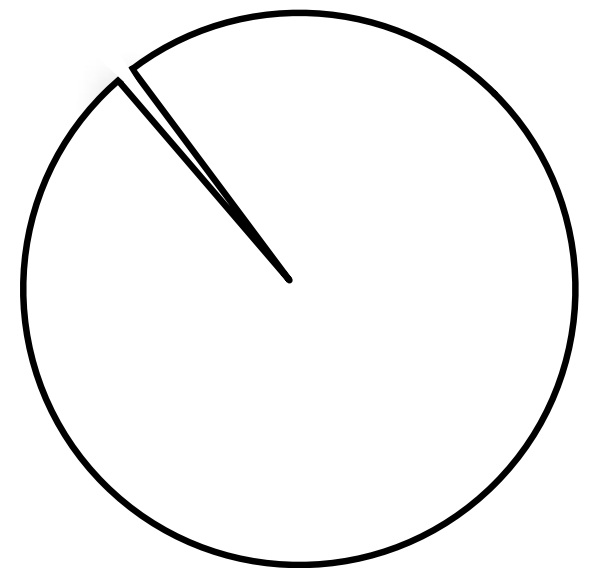
K'_c



$J_c = \partial K_c$



$J'_c = \partial K'_c$



- They are close to each other in the Hausdorff distance

- Their boundaries are not close to each other in the Hausdorff distance